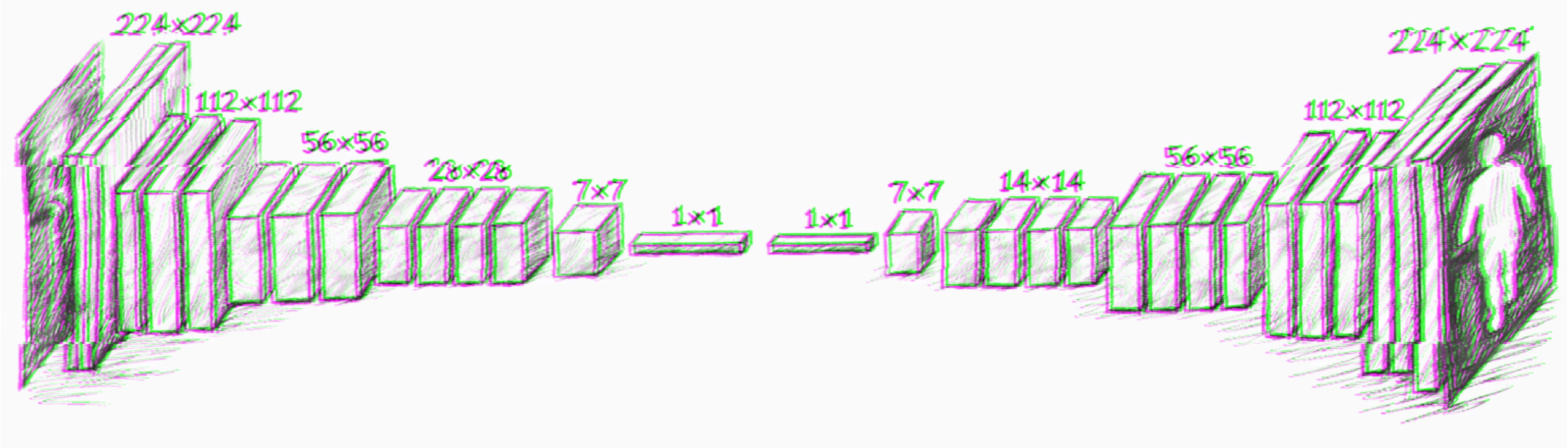
[Tuxae]



[Ateliers Data & ML]/[5. Introduction au Deep Learning]

Introduction

Jupyter-pytorch

The JupyterLab IDE with Python and the deep-learning framework PyTorch.

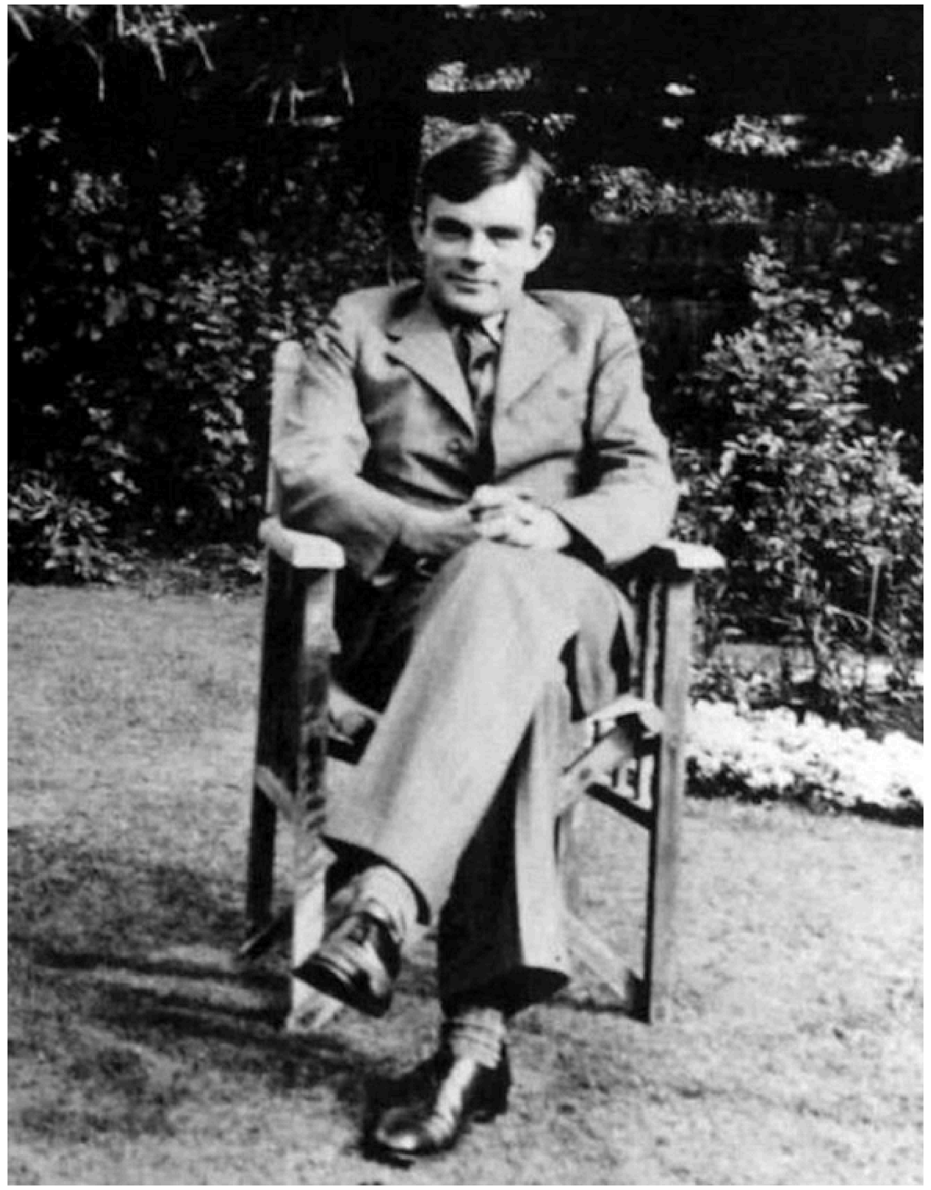
Lancer

https://shorturl.at/wqFL2

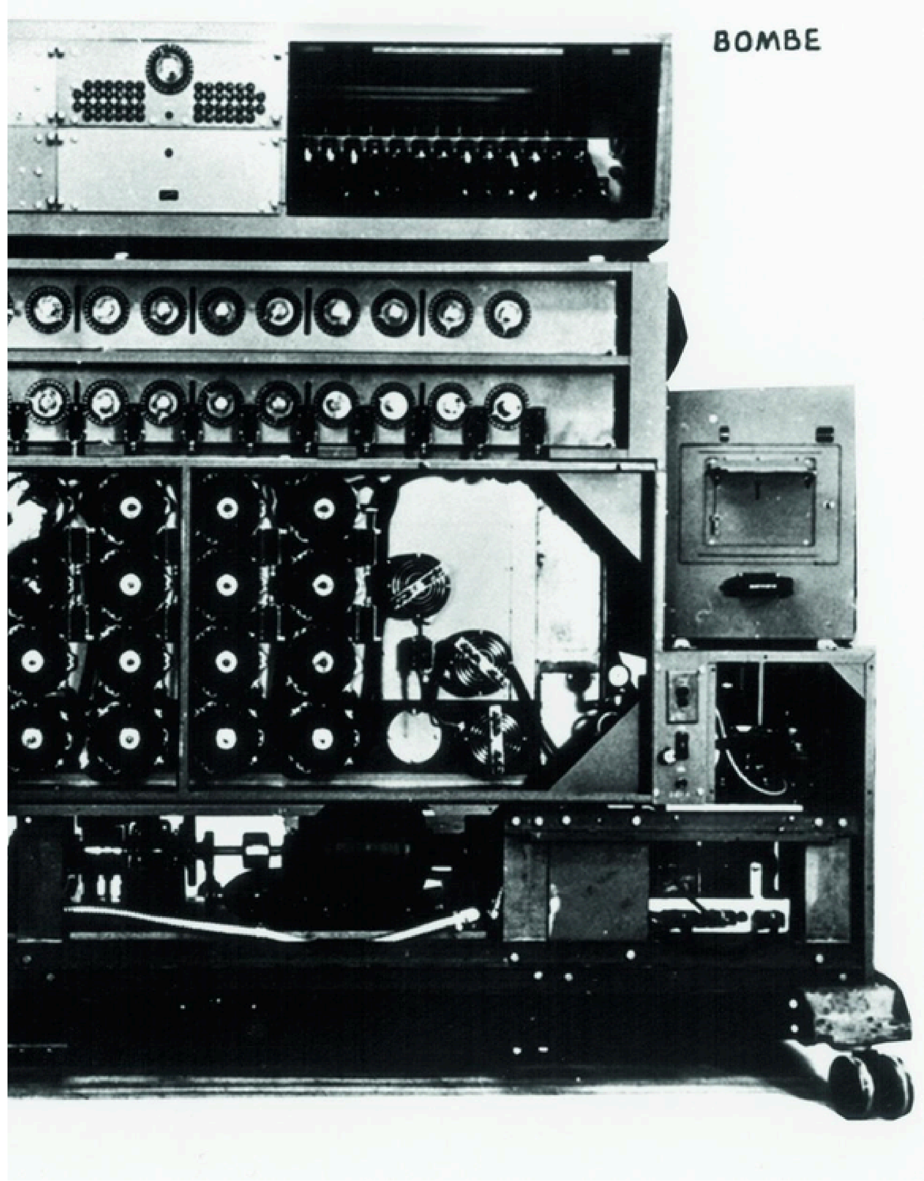
Introduction

– 1943 –
NAISSANCE DU CONNECTIONISME FORMEL

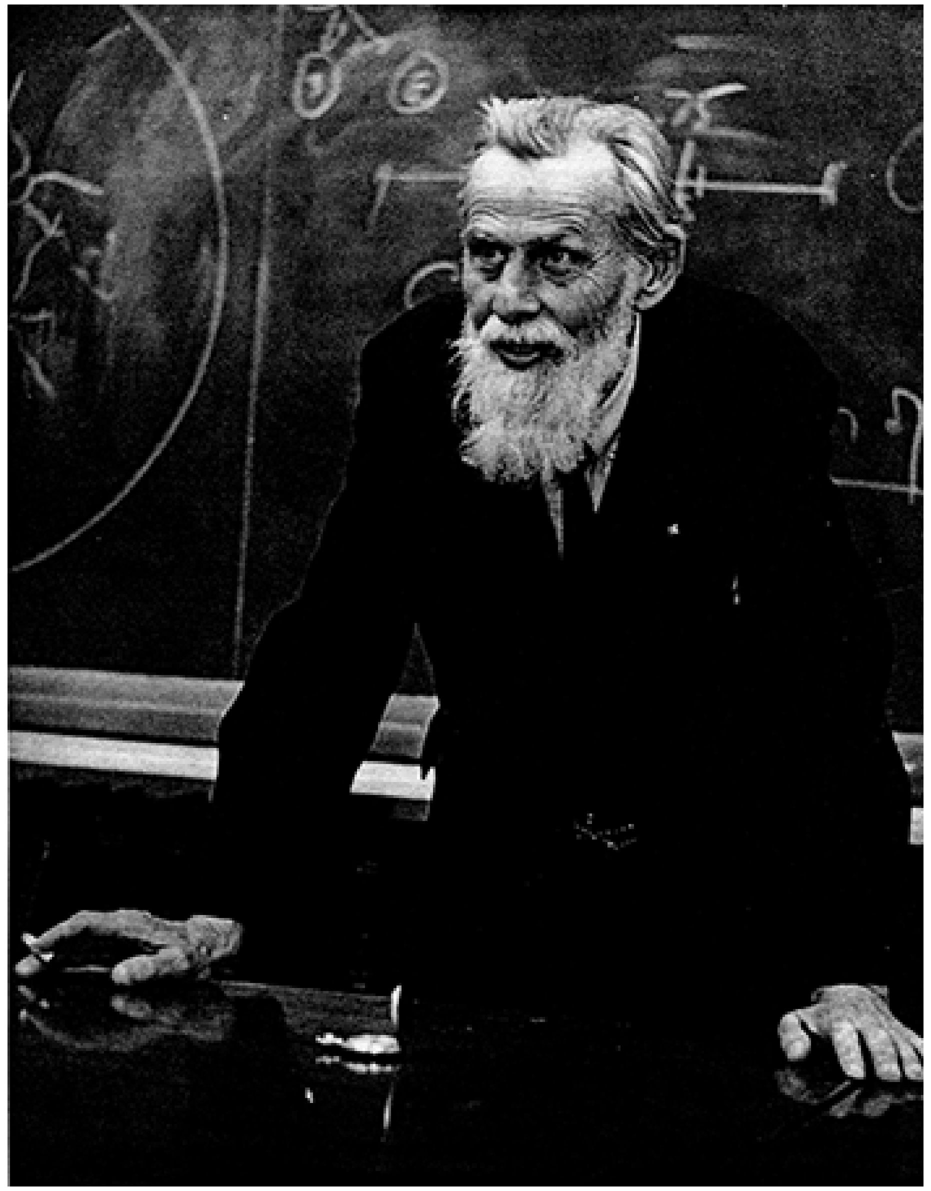
– 1943 –
NAISSANCE DU CONNECTIONISME FORMEL



Alan Turing
Père de l'informatique



– 1943 –
NAISSANCE DU CONNECTIONISME FORMEL



Warren McCulloch
Neurophysiologiste

Walter Pitts
Logicien

Bulletin of Mathematical Biology Vol. 52, No. 1/2, pp. 99–115, 1990.
Printed in Great Britain.

0092-8240/90$3.00 + 0.00
Pergamon Press plc
Society for Mathematical Biology

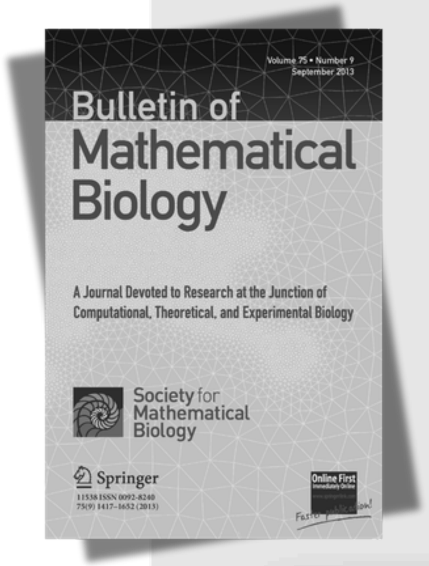
A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY*

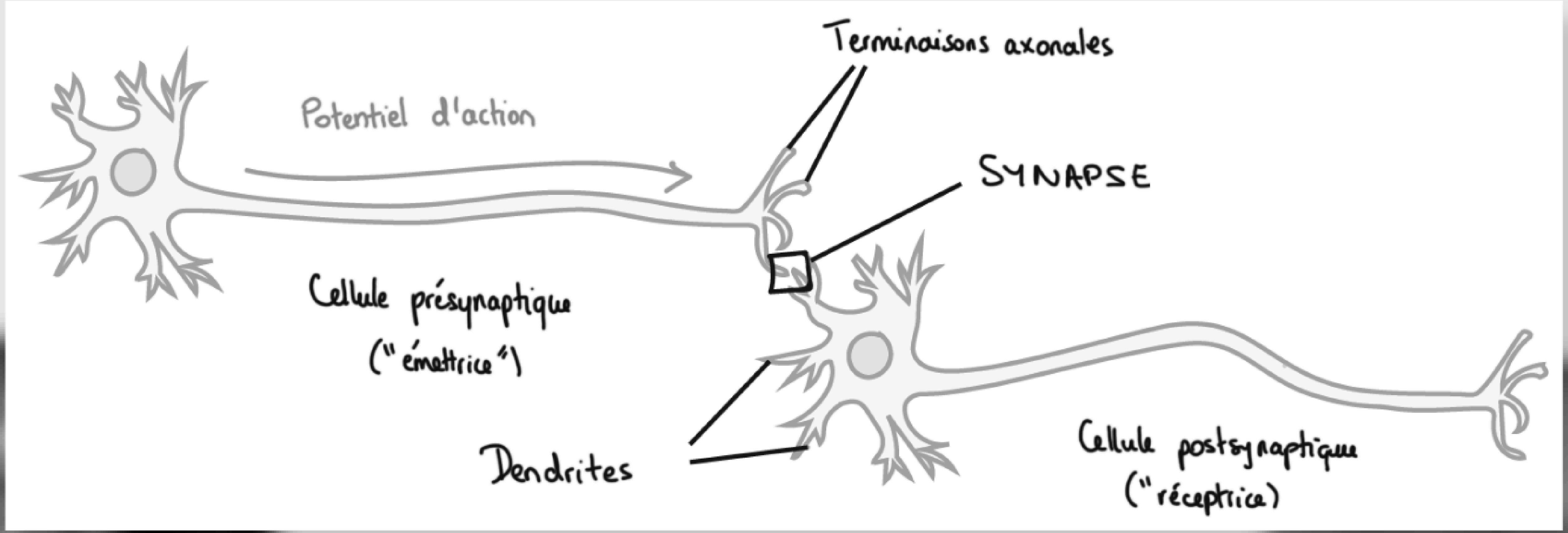
■ WARREN S. MCCULLOCH AND WALTER PITTS
University of Illinois, College of Medicine,
Department of Psychiatry at the Illinois Neuropsychiatric Institute,
University of Chicago, Chicago, U.S.A.

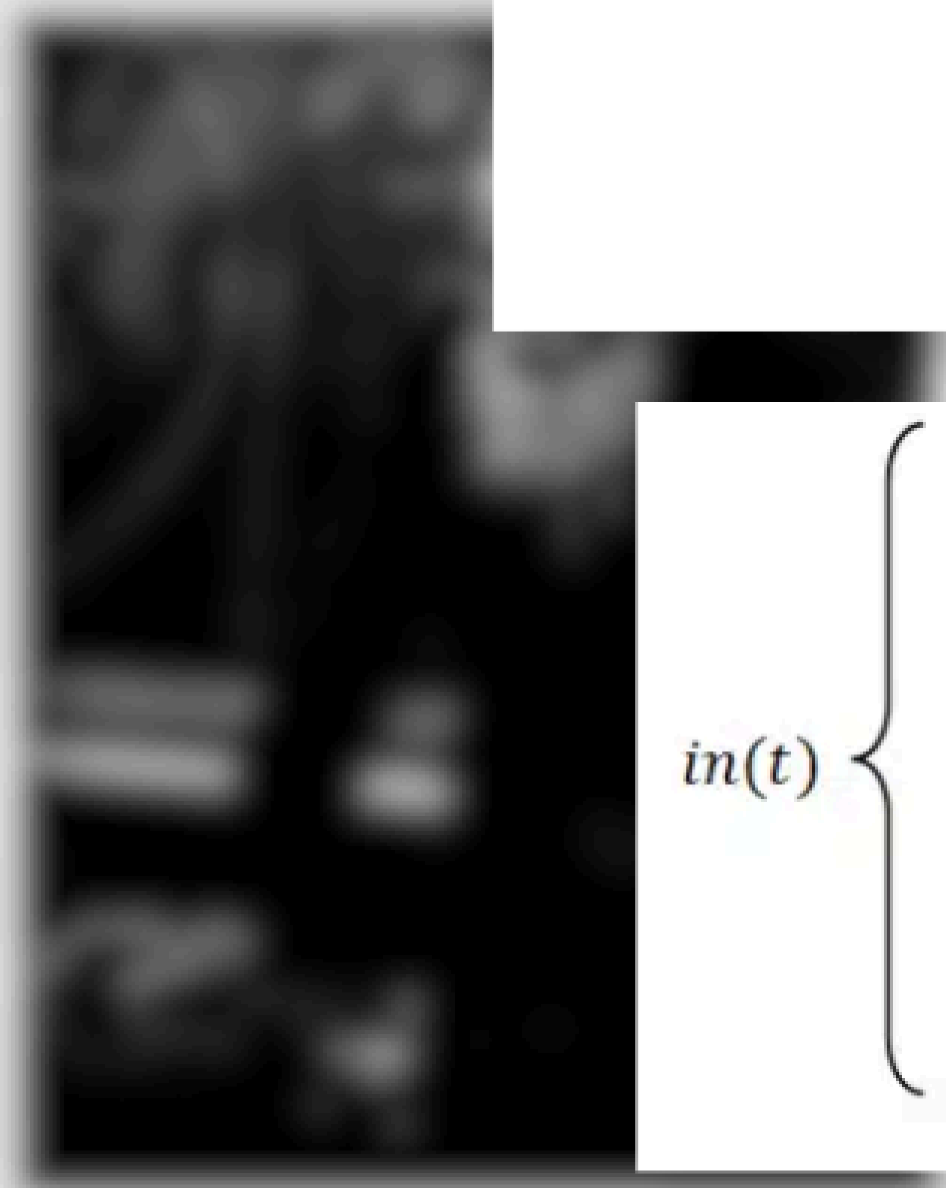
Because of the "all-or-none" character of nervous activity, neural events and the relations among them can be treated by means of propositional logic. It is found that the behavior of every net can be described in these terms, with the addition of more complicated logical means for nets containing circles; and that for any logical expression satisfying certain conditions, one can find a net behaving in the fashion it describes. It is shown that many particular choices among possible neurophysiological assumptions are equivalent, in the sense that for every net behaving under one assumption, there exists another net which behaves under the other and gives the same results, although perhaps not in the same time. Various applications of the calculus are discussed.

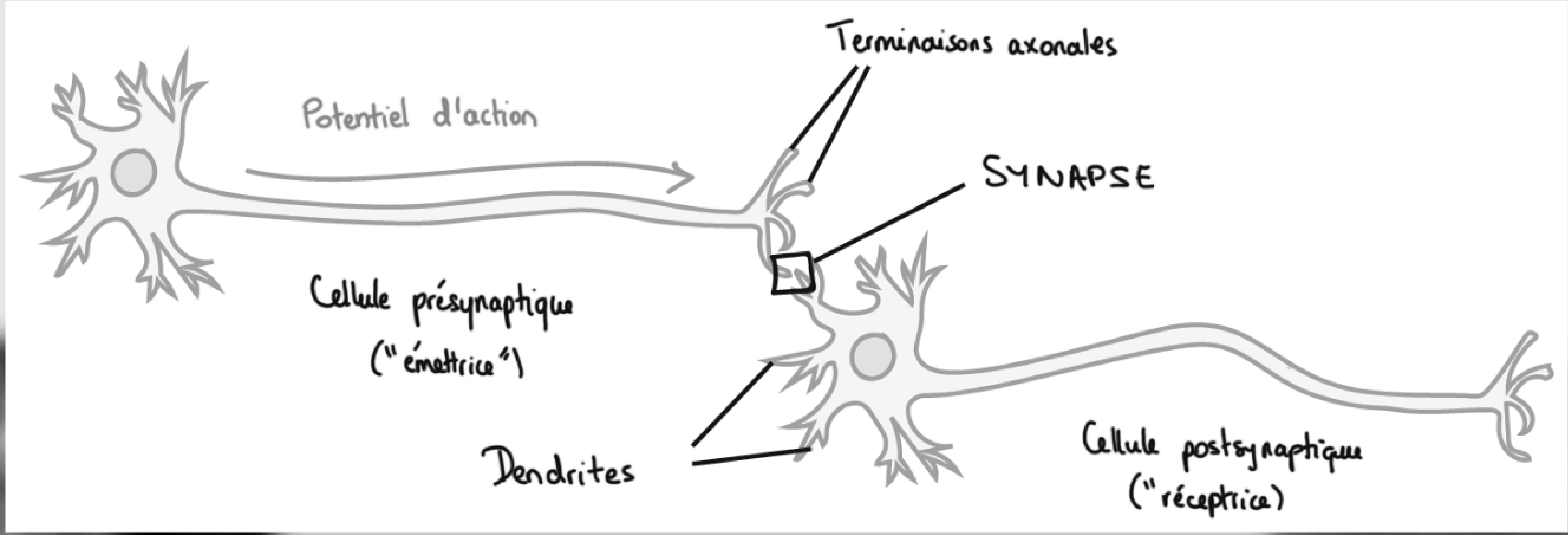
1. Introduction. Theoretical neurophysiology rests on certain cardinal assumptions. The nervous system is a net of neurons, each having a soma and an axon. Their adjunctions, or synapses, are always between the axon of one neuron and the soma of another. At any instant a neuron has some threshold, which excitation must exceed to initiate an impulse. This, except for the fact and the time of its occurence, is determined by the neuron, not by the excitation. From the point of excitation the impulse is propagated to all parts of the neuron. The velocity along the axon varies directly with its diameter, from $<1\ \mathrm{ms}^{-1}$ in thin axons, which are usually short, to $>150\ \mathrm{ms}^{-1}$ in thick axons, which are usually long. The time for axonal conduction is consequently of little importance in determining the time of arrival of impulses at points unequally remote from the same source. Excitation across synapses occurs predominantly from axonal terminations to somata. It is still a moot point whether this depends upon irreciprocity of individual synapses or merely upon prevalent anatomical configurations. To suppose the latter requires no hypothesis *ad hoc* and explains known exceptions, but any assumption as to cause is compatible with the calculus to come. No case is known in which excitation through a single synapse has elicited a nervous impulse in any neuron, whereas any neuron may be excited by impulses arriving at a sufficient number of neighboring synapses within the period of latent addition, which lasts <0.25 ms. Observed temporal summation of impulses at greater intervals

* Reprinted from the *Bulletin of Mathematical Biophysics*, Vol. 5, pp. 115–133 (1943).

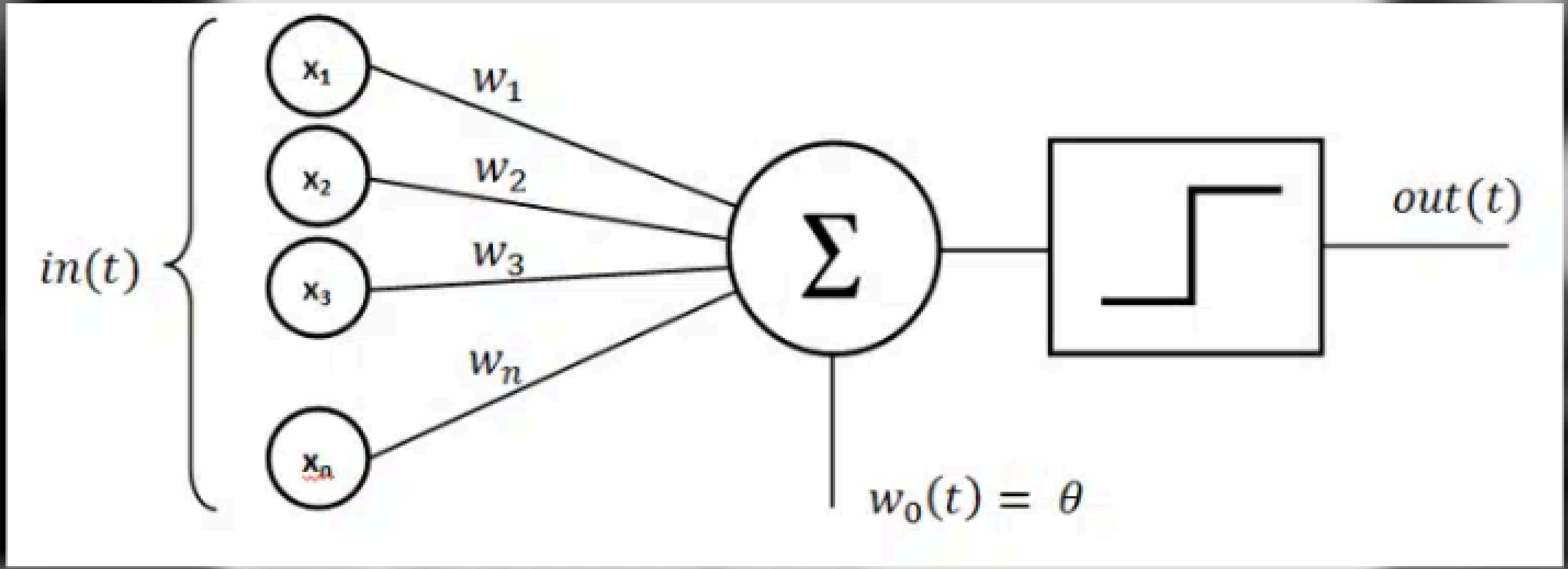




Terminaisons axonales
Potentiel d'action
SYNAPSE
Cellule présynaptique
("émettrice")
Dendrites
Cellule postsynaptique
("réceptrice")

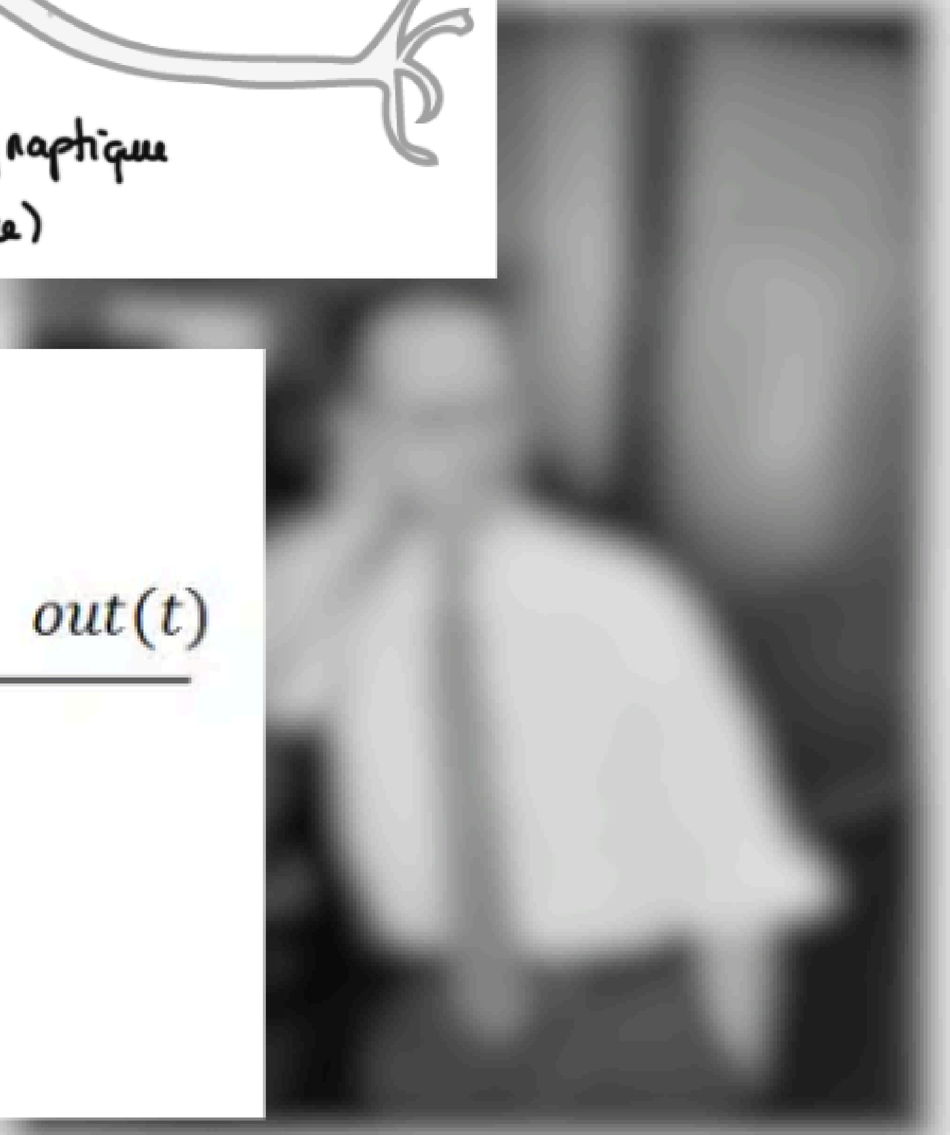





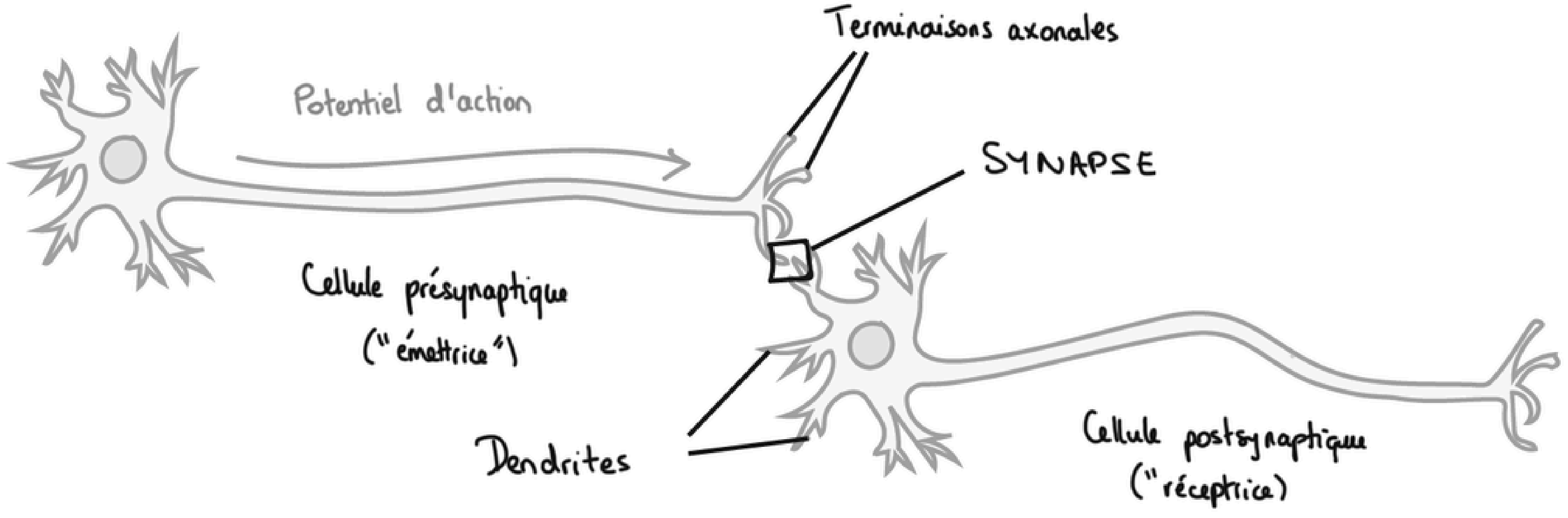
Terminaisons axonales
Potentiel d'action
SYNAPSE
Cellule présynaptique
("émettrice")
Dendrites
Cellule postsynaptique
("réceptrice")

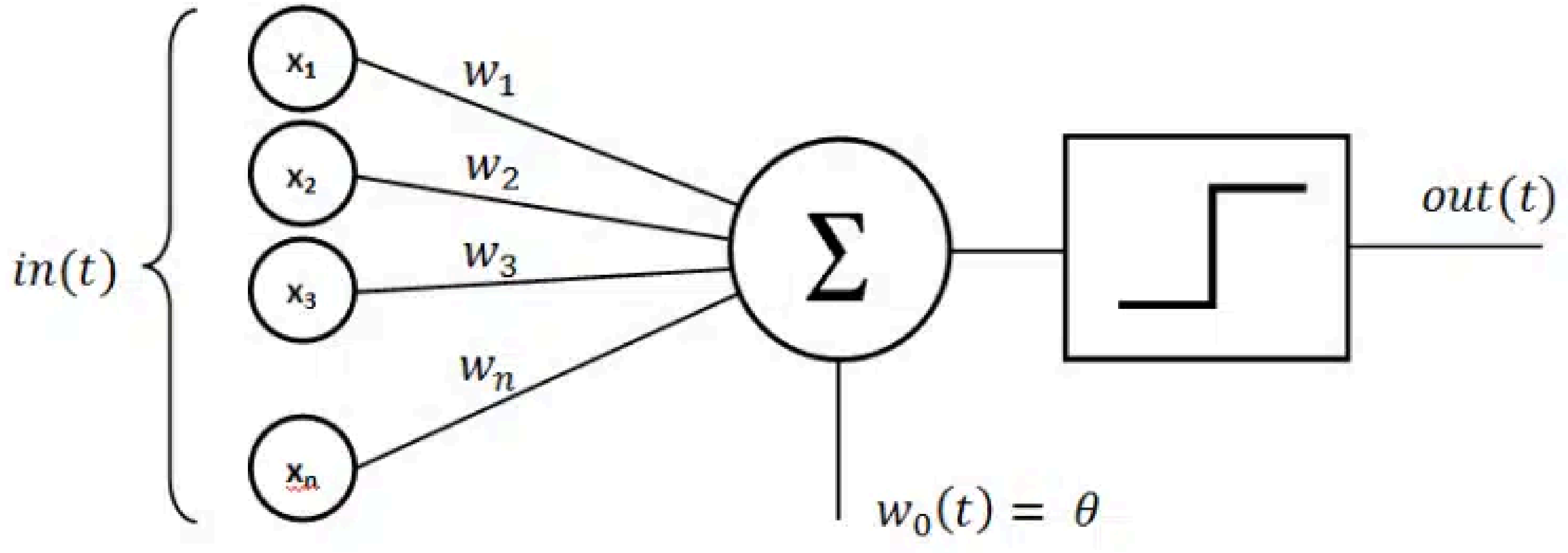


in(t)
x_1
x_2
x_3
x_n
w_1
w_2
w_3
w_n
Σ
out(t)
$w_0(t) = \theta$



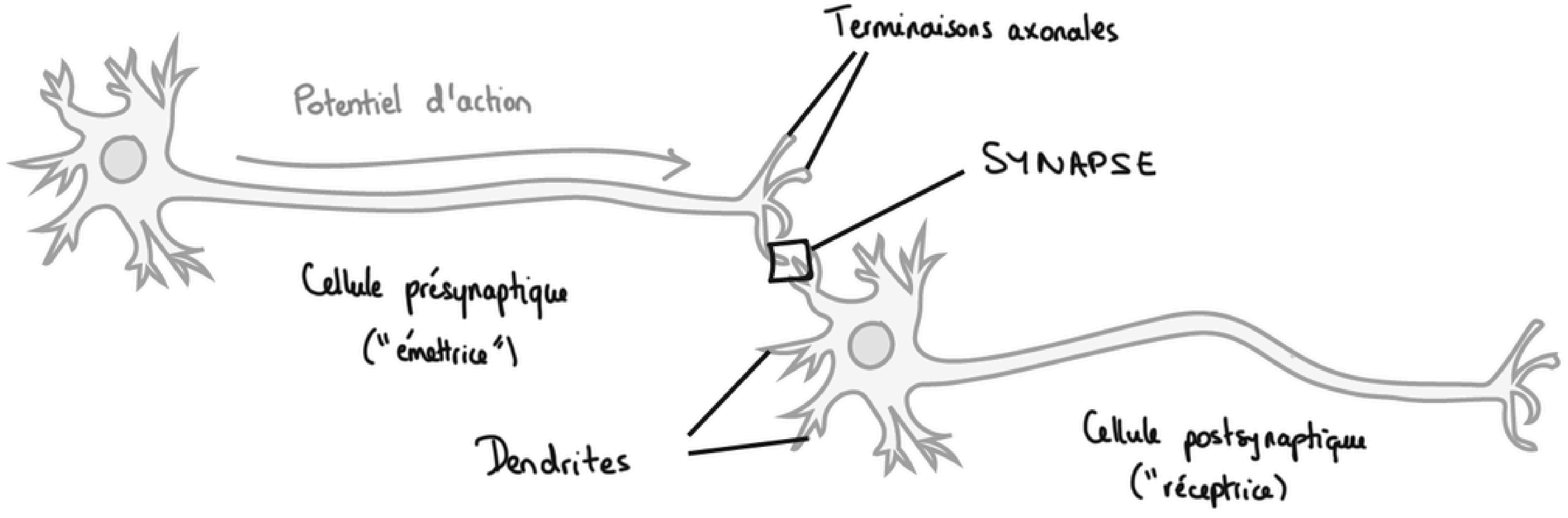
– 1943 –
NAISSANCE DU CONNECTIONISME FORMEL

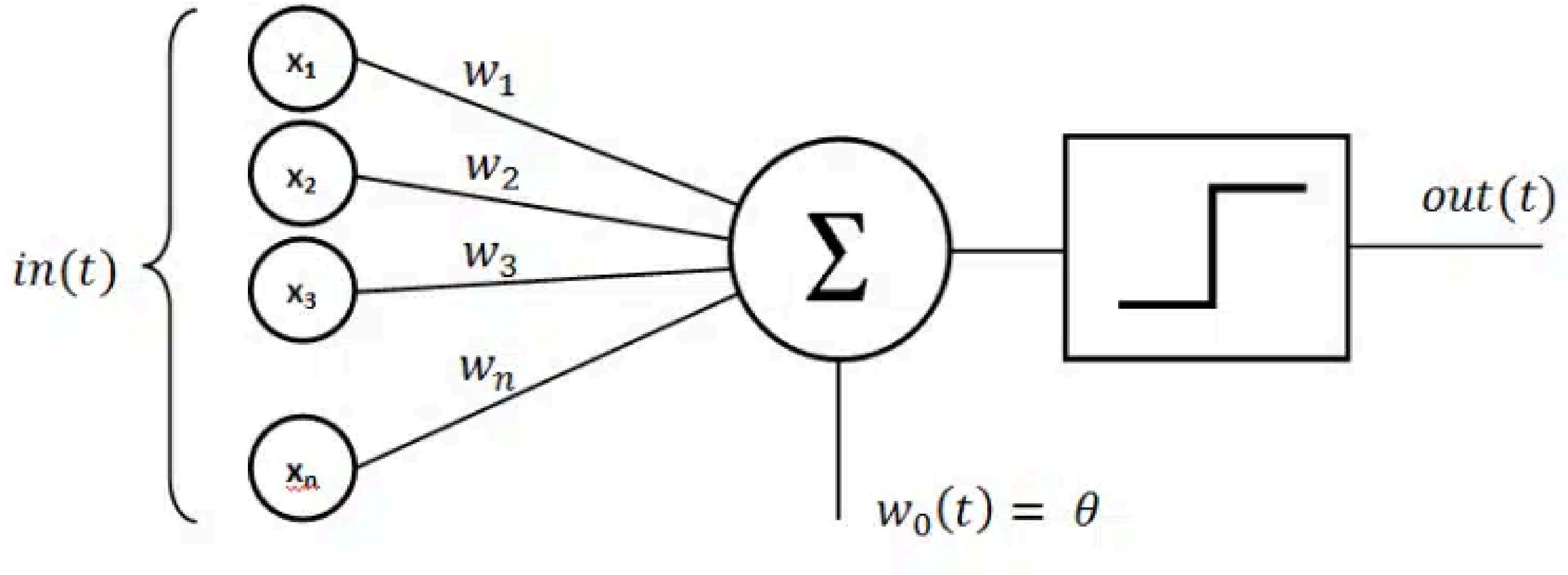




$$\hat{y} = \mathbb{1}\left(\sum_k \omega_k x_k > b\right)$$

– 1943 –
NAISSANCE DU CONNECTIONISME FORMEL

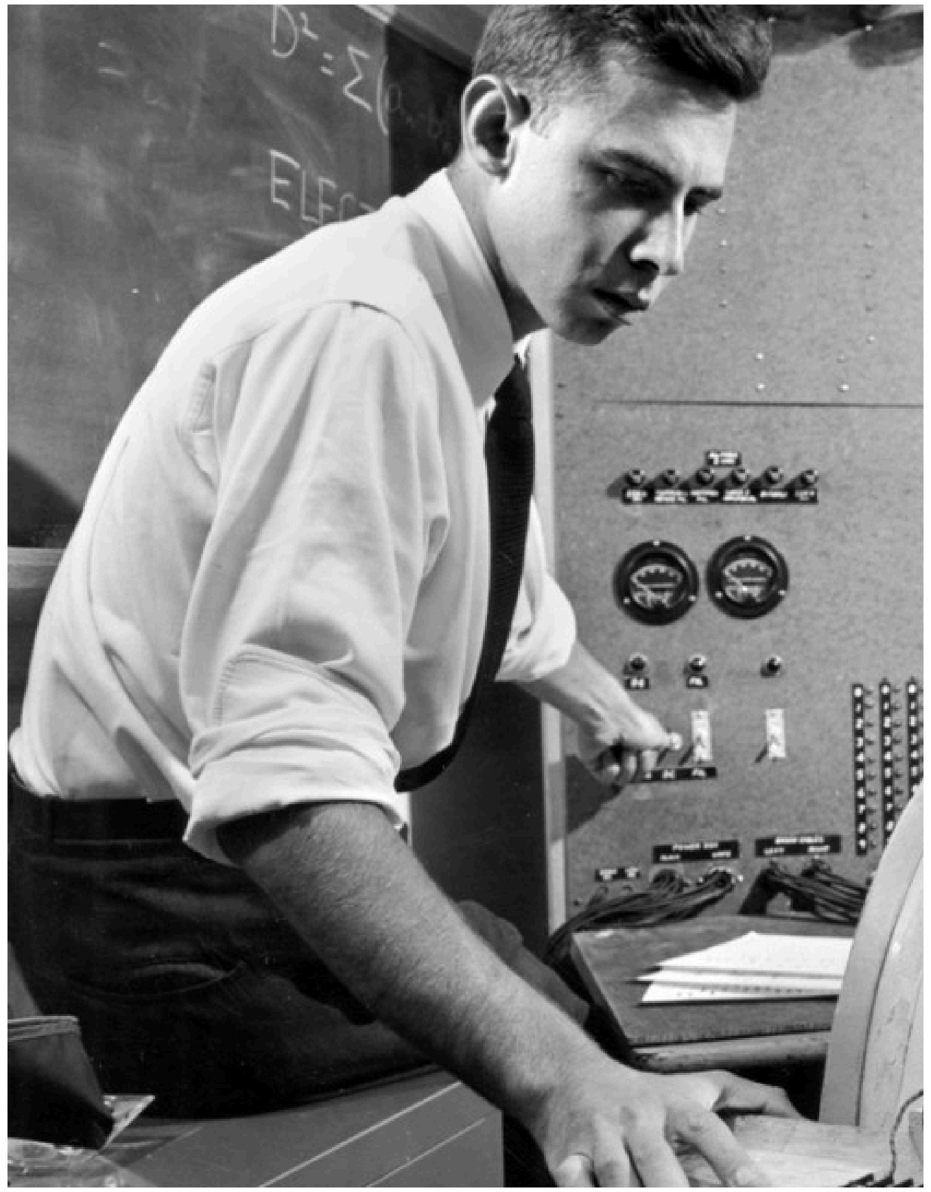




$$\hat{y} = \mathbb{1}\left(\sum_k \omega_k x_k > b\right)$$

$$\hat{y} = \operatorname{sgn}\left(\omega^\top x - b\right)$$

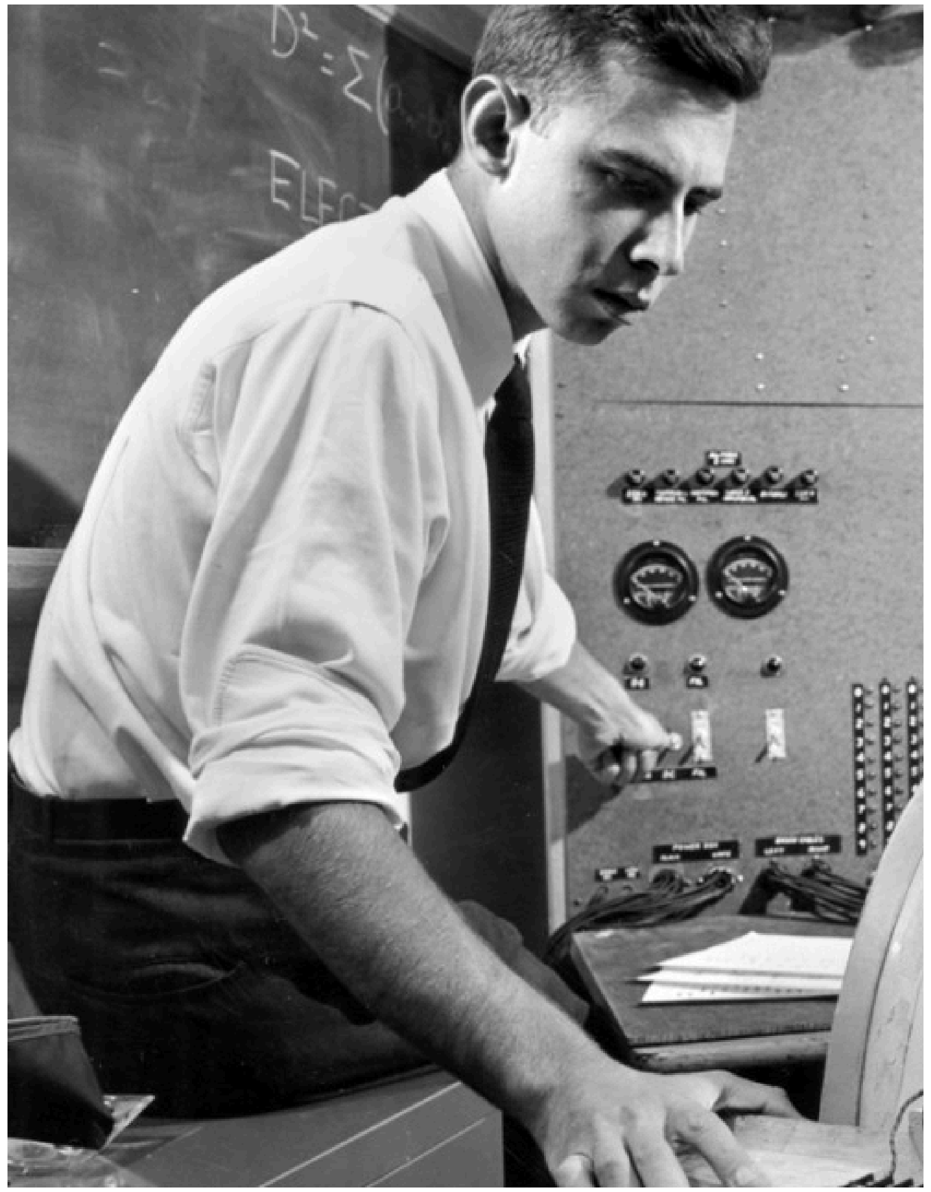
– 1958 –
LE PERCEPTRON



Mark I Perceptron

Frank Rosenblatt
Psychologue et informaticien

– 1958 –
LE PERCEPTRON

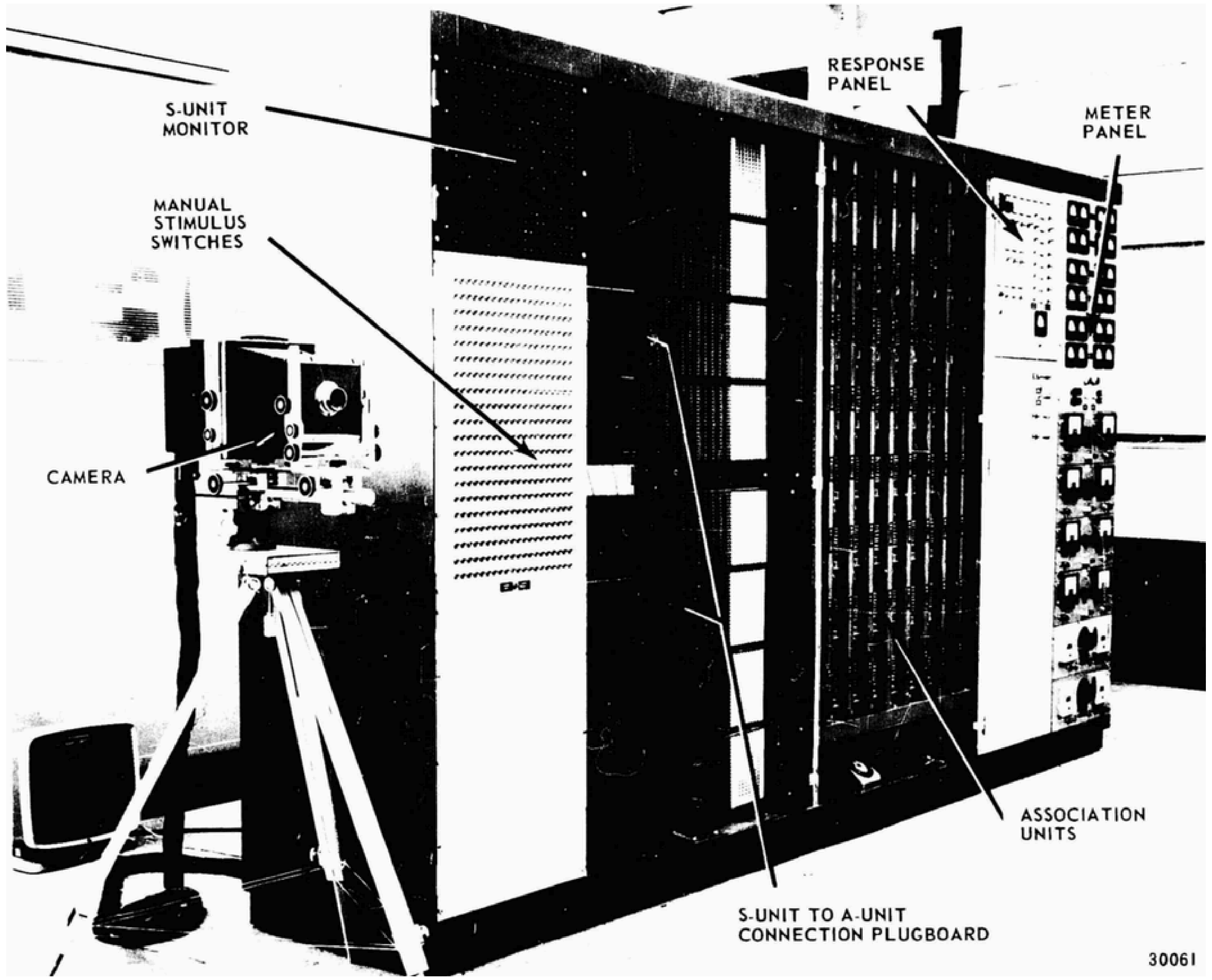


Frank Rosenblatt
Psychologue et informaticien

$$\hat{y}_i = \operatorname{sgn}\left(\omega^\top x_i - b\right)$$

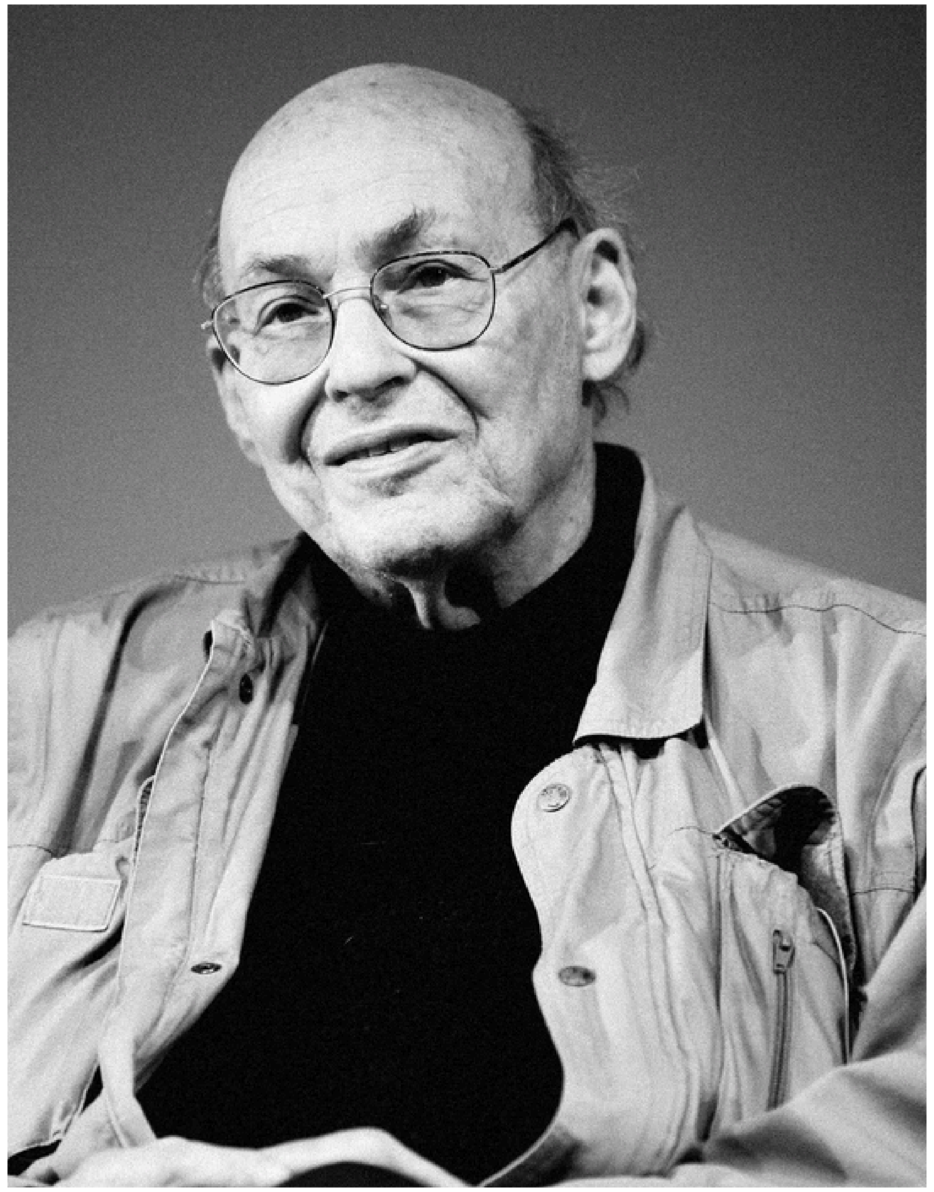
$$\omega \leftarrow 0$$
$$\omega \leftarrow \omega + \eta(y_i - \hat{y}_i)x$$

– 1958 –
LE PERCEPTRON



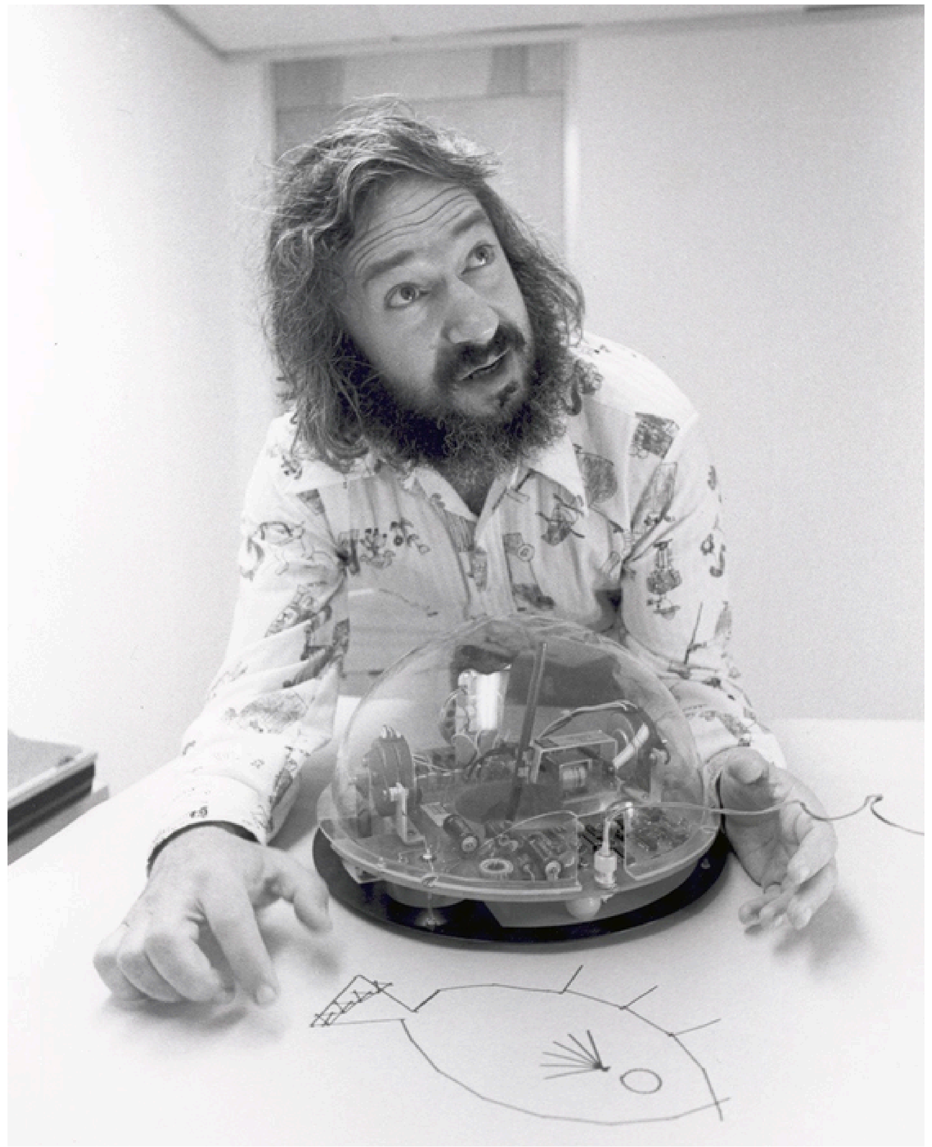
Mark I Perceptron

– 1969 –
AI WINTER

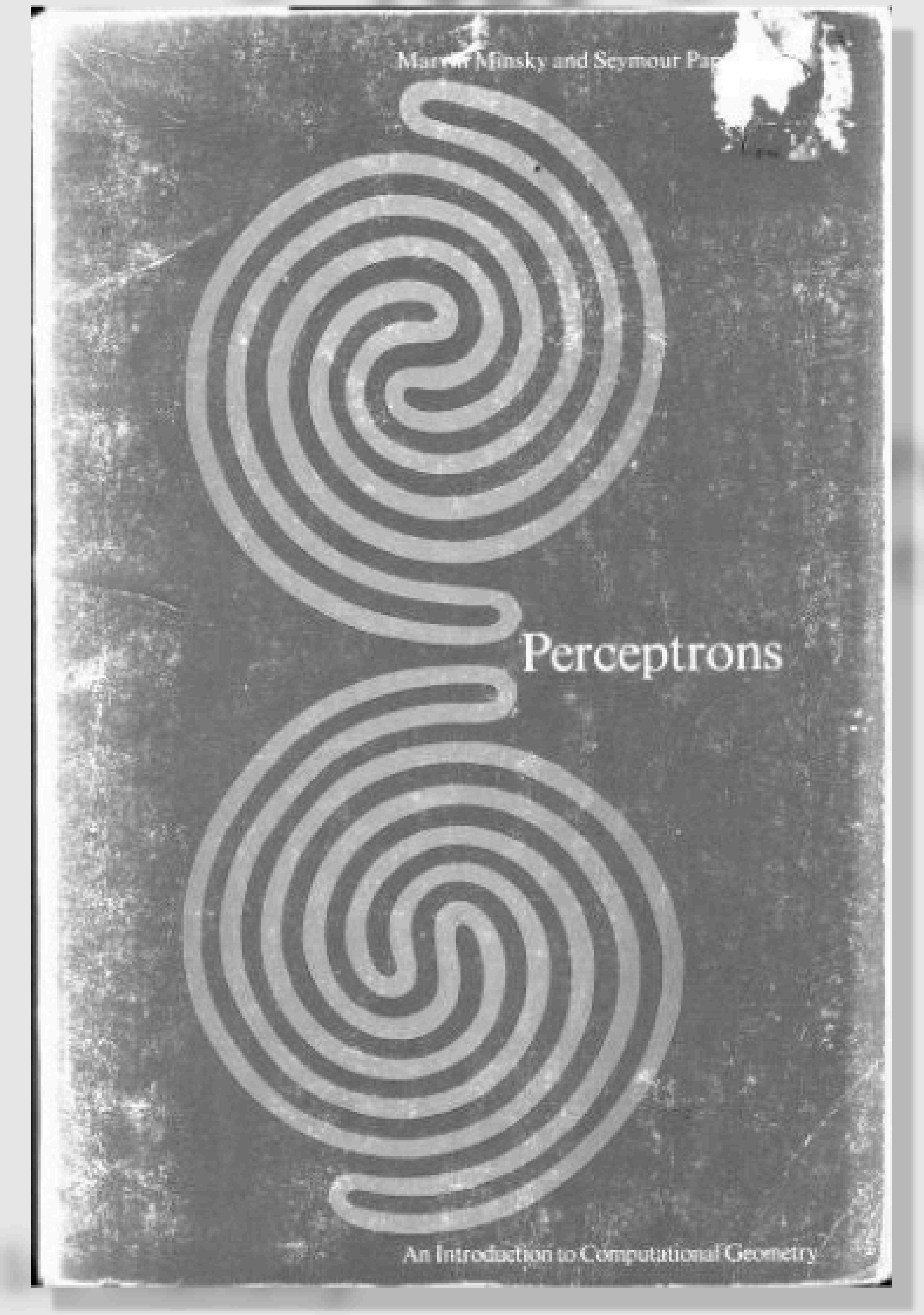


Marvin Minsky
Mathématicien

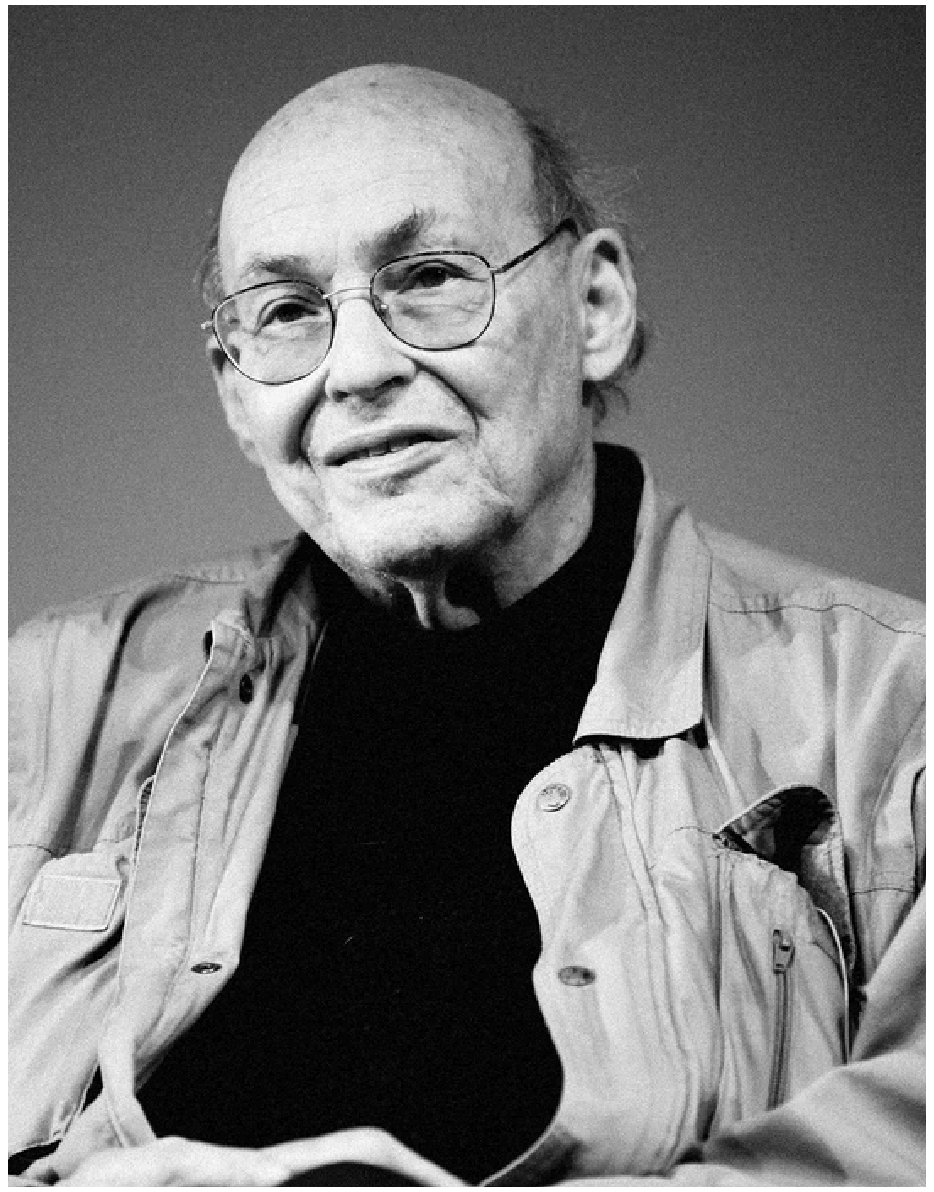
Seymour Papert
Mathématicien



- 1969 -

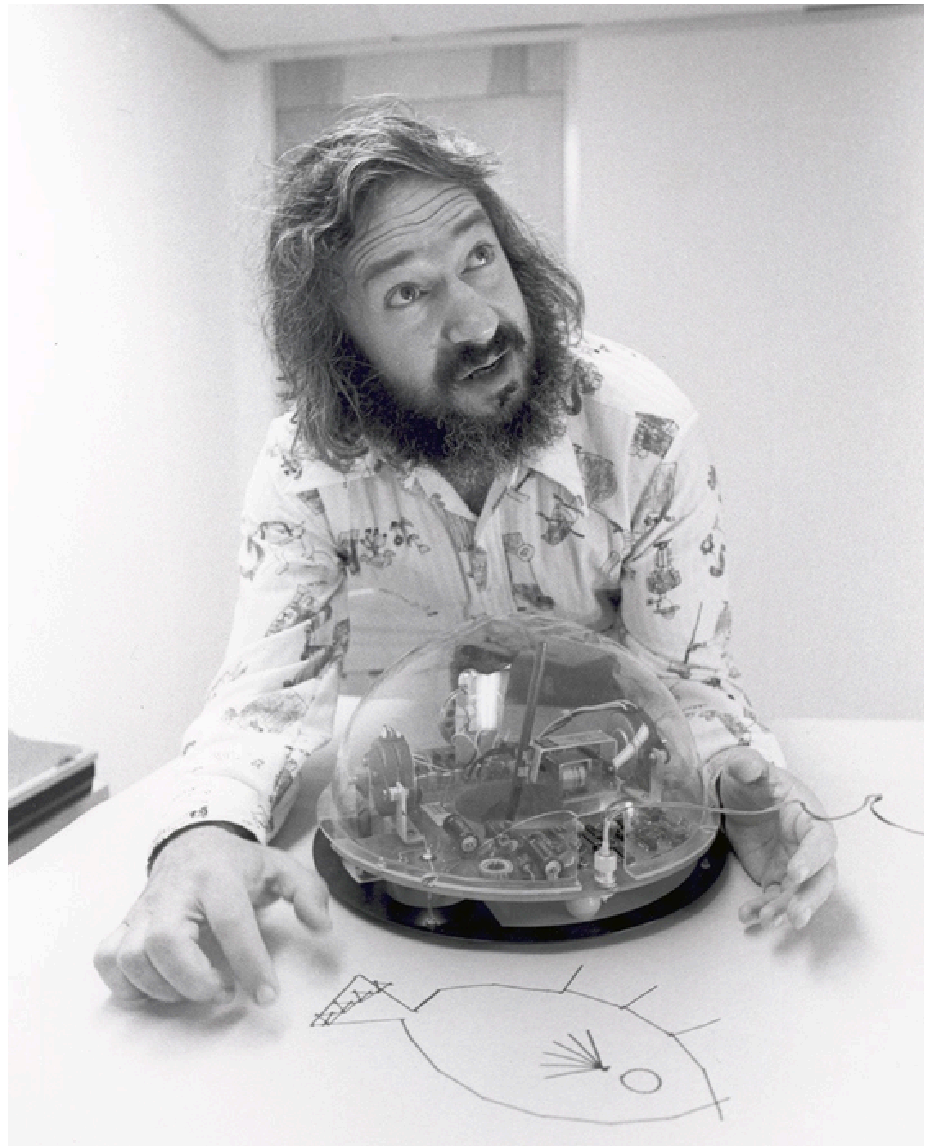

Marvin Minsky and Seymour Pap
Perceptrons
An Introduction to Computational Geometry

– 1969 –
AI WINTER

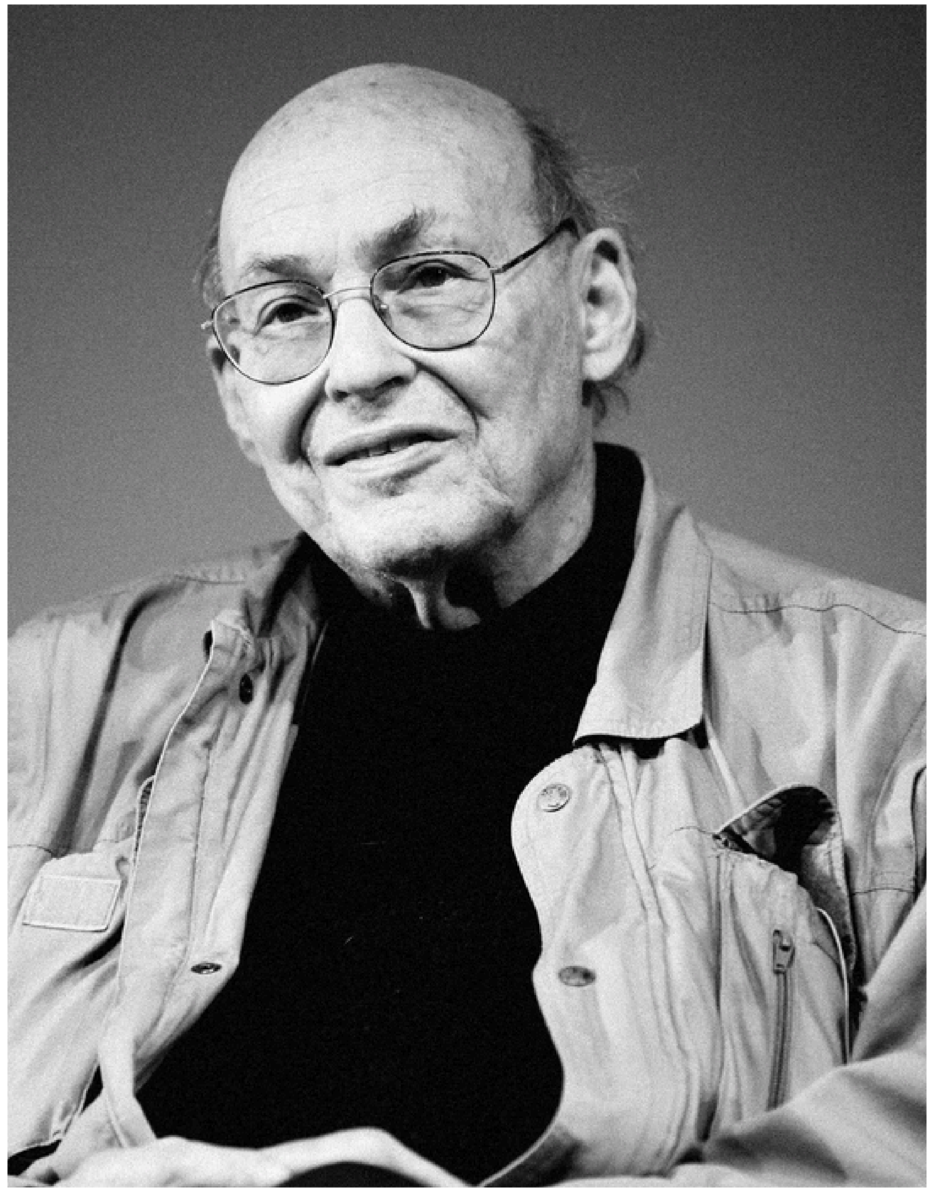


Marvin Minsky
Mathématicien

Seymour Papert
Mathématicien

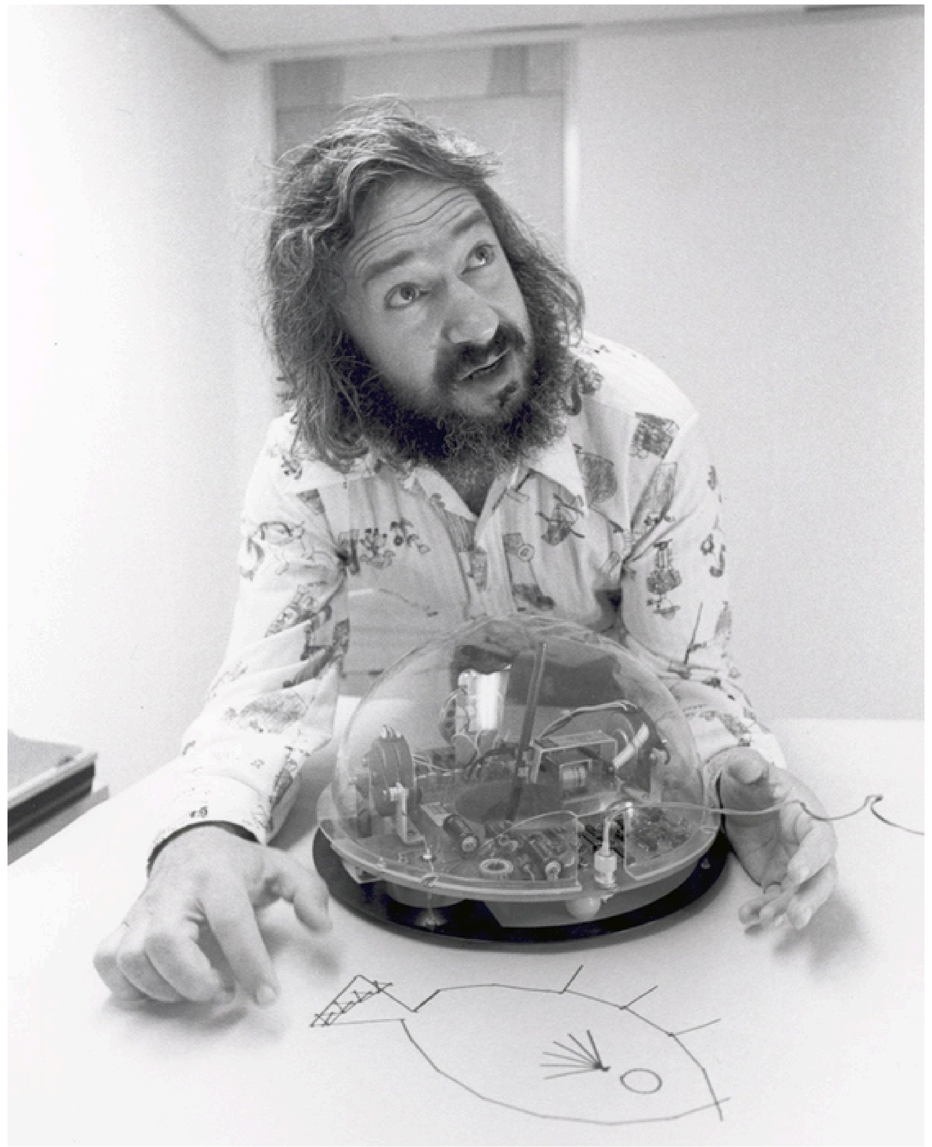


– 1969 –
AI WINTER

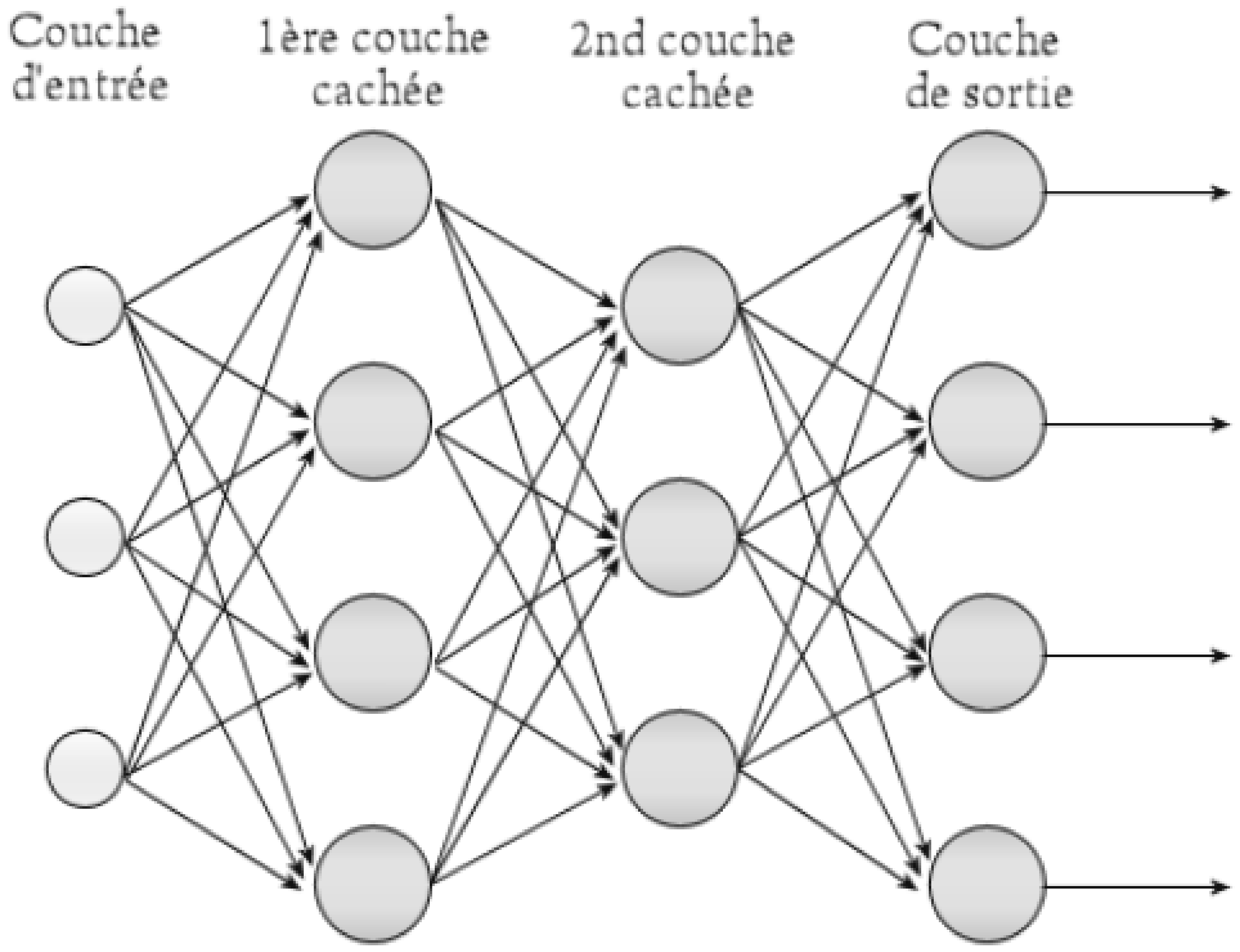


Marvin Minsky
Mathématicien

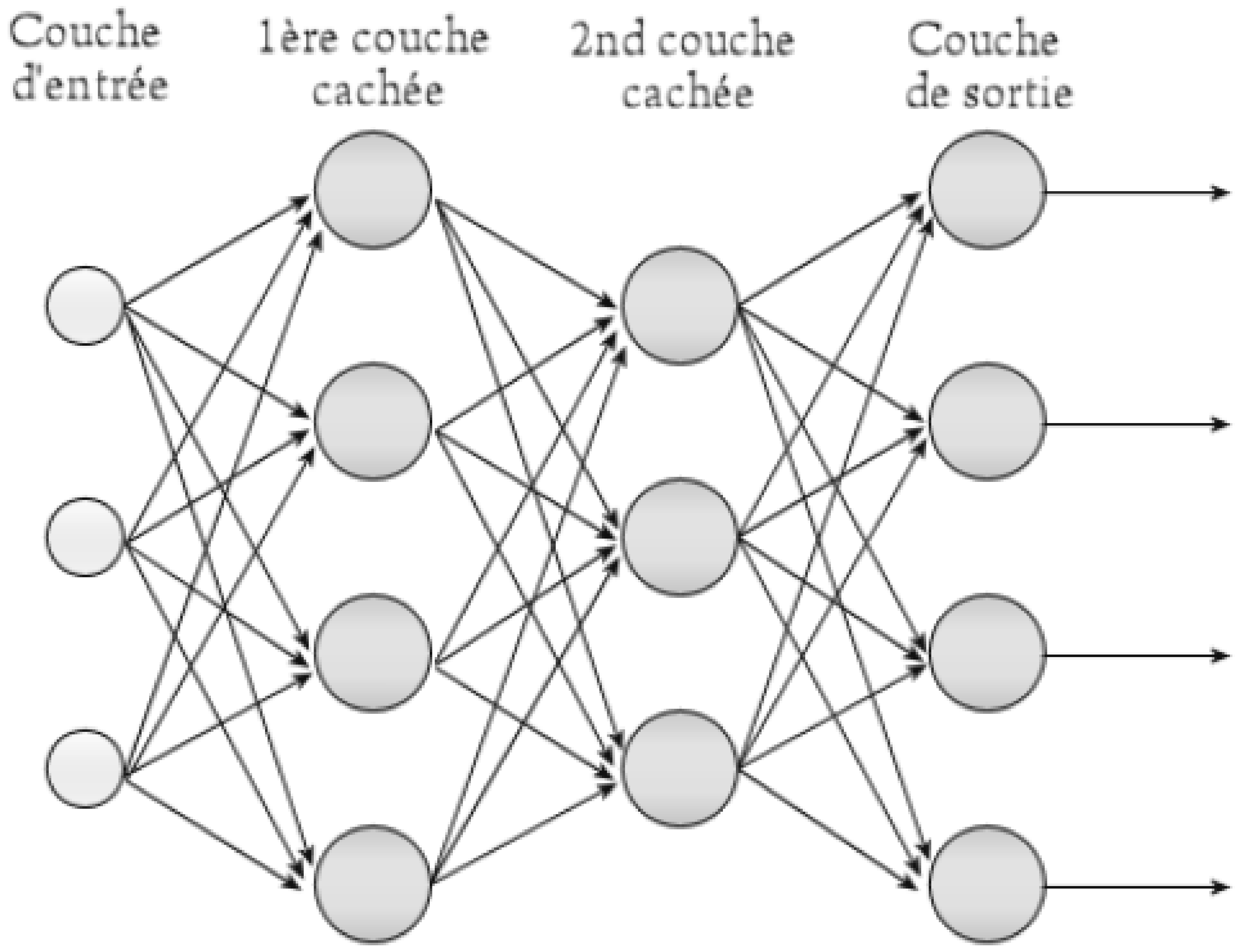
Seymour Papert
Mathématicien



– 1980 –
MULTI LAYER PERCEPTRON

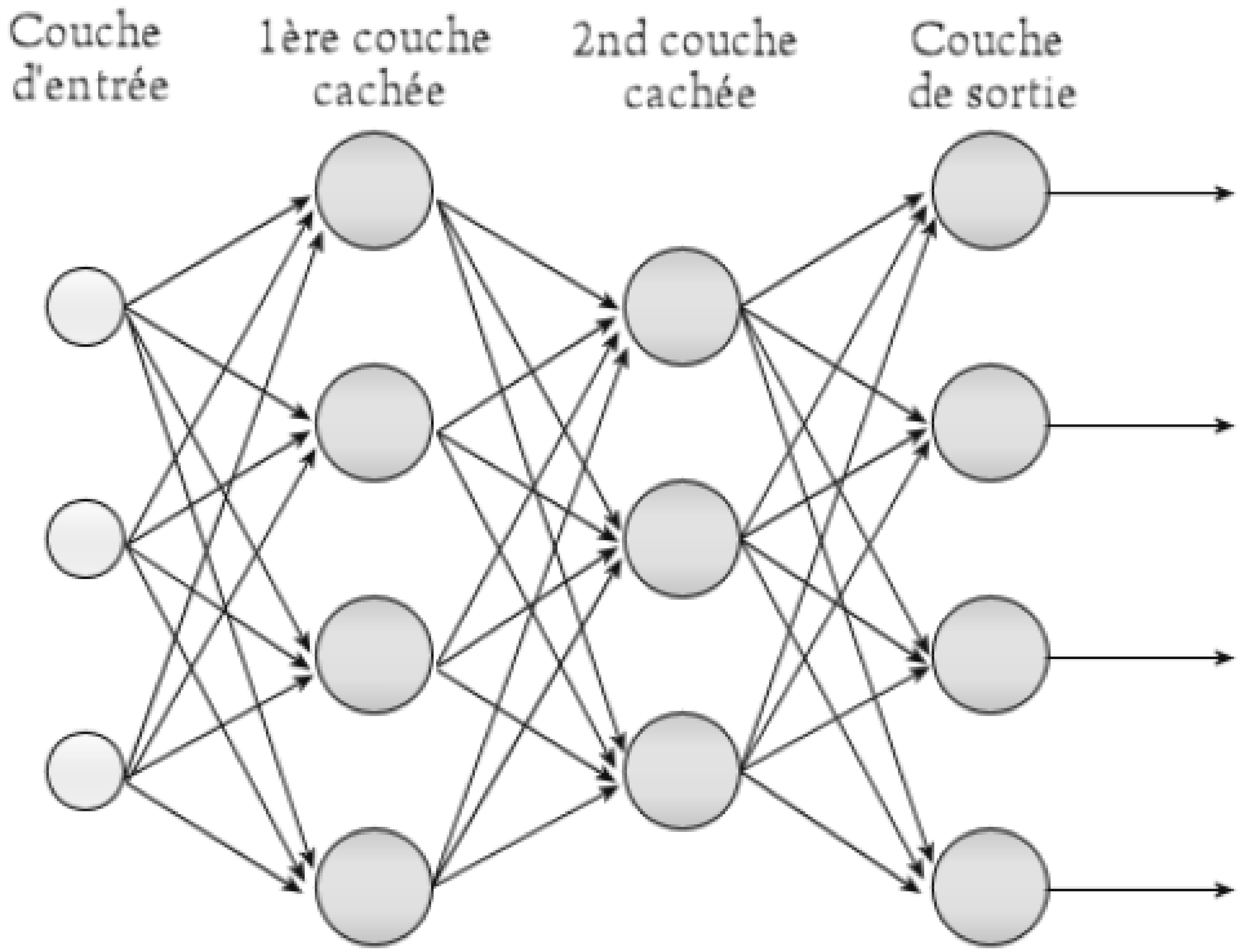


– 1980 –
MULTI LAYER PERCEPTRON



$$x_1 = \texttt{entree}$$
$$x_2 = \sigma\left(W_1 x_1 + b_1\right)$$
$$x_3 = \sigma\left(W_2 x_2 + b_2\right)$$
$$\vdots$$
$$y = x_n$$

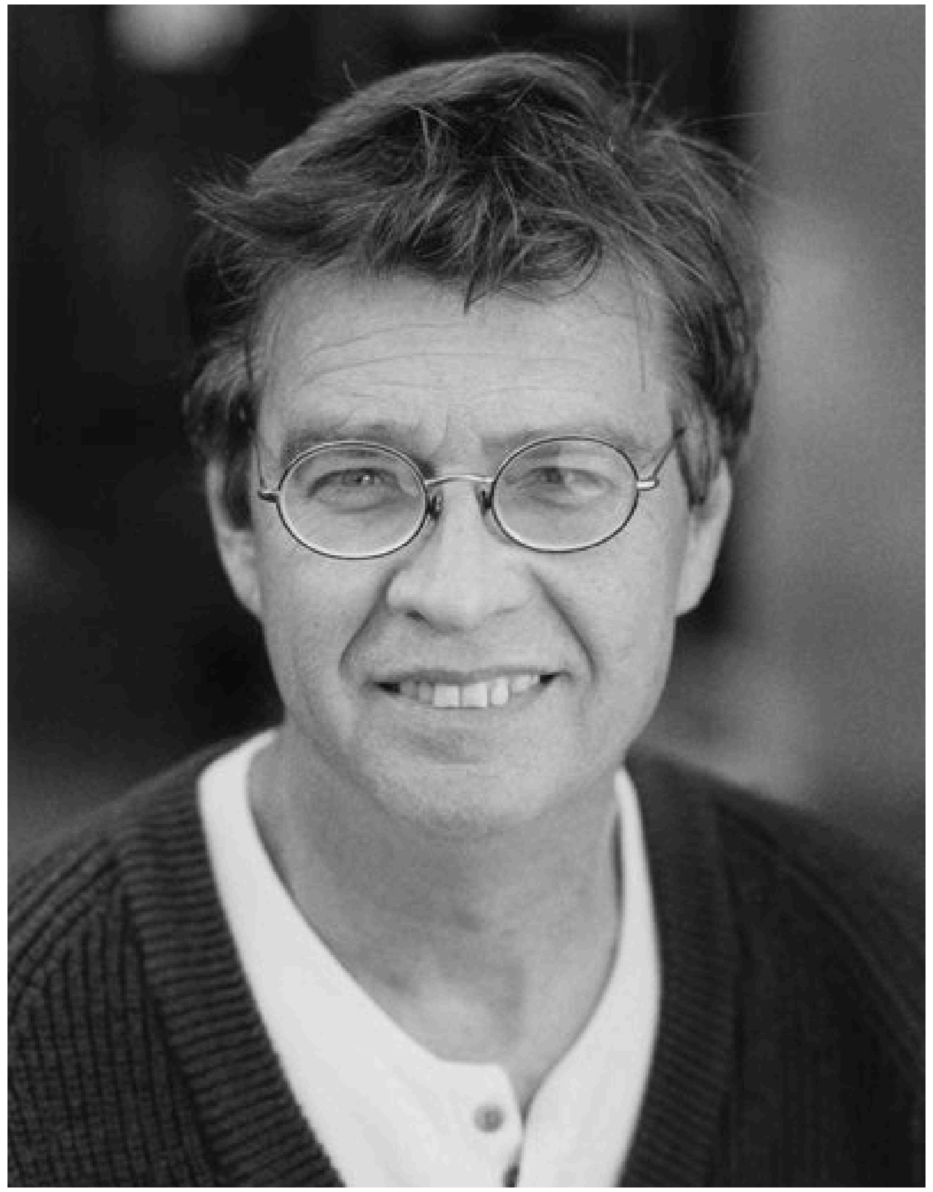
- 1980 -
MULTI LAYER PERCEPTRON



$$x_1 = \texttt{entree}$$
$$x_2 = \sigma\left(W_1 x_1 + b_1\right)$$
$$x_3 = \sigma\left(W_2 x_2 + b_2\right)$$
$$\vdots$$
$$y = x_n$$

Théorème d'approximation universelle

– 1986 –
BACK PROPAGATION



Geoffrey Hinton
Mathématicien et informaticien
Prix Turing 2018
Prix Nobel (Physique) 2024

David Rumelhart
Mathématicien

– 1986 –

BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$

$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$

$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$

$$\vdots$$

$$\hat{y}_i = x_{i,n}$$

– 1986 –
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$
$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$
$$\vdots$$
$$\hat{y}_i = x_{i,n}$$

$$\ell\left(y, \hat{y}\right) = \sum_i \left(y_i - \hat{y}_i\right)^2 \quad \texttt{(MSE)}$$

– 1986 –
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$
$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$
$$\vdots$$
$$\hat{y}_i = x_{i,n}$$

$$\ell\left(y, \hat{y}\right) = \sum_i \left(y_i - \hat{y}_i\right)^2 \quad \texttt{(MSE)}$$
$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$

– 1986 –
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$
$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$
$$\vdots$$
$$\hat{y}_i = x_{i,n}$$

$$\ell\left(y, \hat{y}\right) = \sum_i \left(y_i - \hat{y}_i\right)^2 \quad \texttt{(MSE)}$$
$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$
$$\theta \leftarrow \theta - \eta \sum_i \frac{\partial \ell_i}{\partial \theta}$$

– 1986 –
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$
$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$
$$\vdots$$
$$\hat{y}_i = x_{i,n}$$

$$\ell\left(y, \hat{y}\right) = \sum_i \left(y_i - \hat{y}_i\right)^2 \quad \texttt{(MSE)}$$
$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$
$$\theta \leftarrow \theta - \eta \sum_i \frac{\partial \ell_i}{\partial \theta}$$
$$\frac{\partial \ell_i}{\partial W_{j-1}} \qquad \frac{\partial \ell_i}{\partial b_{j-1}}$$

– 1986 –
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$

$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$

$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$

$$\vdots$$

$$\hat{y}_i = x_{i,n}$$

$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$

$$\frac{\partial \ell_i}{\partial W_{j-1}} \qquad \frac{\partial \ell_i}{\partial b_{j-1}}$$

– 1986 –
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$
$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$
$$\vdots$$
$$\hat{y}_i = x_{i,n}$$
$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$

$$a_{i,j} = W_{j-1} x_{i,j-1} + b_{j-1}, \qquad x_{i,j} = \sigma(a_{i,j})$$

$$\frac{\partial \ell_i}{\partial W_{j-1}} \qquad \frac{\partial \ell_i}{\partial b_{j-1}}$$

– 1986 –
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$
$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$
$$\vdots$$
$$\hat{y}_i = x_{i,n}$$
$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$

$$a_{i,j} = W_{j-1} x_{i,j-1} + b_{j-1}, \qquad x_{i,j} = \sigma(a_{i,j})$$

$$\delta_{i,j} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,j}}$$

$$\frac{\partial \ell_i}{\partial W_{j-1}} \qquad \frac{\partial \ell_i}{\partial b_{j-1}}$$

– 1986 –
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$
$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$
$$\vdots$$
$$\hat{y}_i = x_{i,n}$$
$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$

$$a_{i,j} = W_{j-1} x_{i,j-1} + b_{j-1}, \qquad x_{i,j} = \sigma(a_{i,j})$$

$$\delta_{i,j} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,j}}$$

$$\delta_{i,n} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,n}}$$

$$\frac{\partial \ell_i}{\partial W_{j-1}} \qquad \frac{\partial \ell_i}{\partial b_{j-1}}$$

- 1986 -
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$
$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$
$$\vdots$$
$$\hat{y}_i = x_{i,n}$$
$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$

$$\frac{\partial \ell_i}{\partial W_{j-1}} \qquad \frac{\partial \ell_i}{\partial b_{j-1}}$$

$$a_{i,j} = W_{j-1} x_{i,j-1} + b_{j-1}, \qquad x_{i,j} = \sigma(a_{i,j})$$

$$\delta_{i,j} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,j}}$$

$$\delta_{i,n} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,n}} = \frac{\partial \ell(y_i, x_{i,n})}{\partial a_{i,n}}$$

– 1986 –
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$
$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$
$$\vdots$$
$$\hat{y}_i = x_{i,n}$$
$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$

$$\frac{\partial \ell_i}{\partial W_{j-1}} \qquad \frac{\partial \ell_i}{\partial b_{j-1}}$$

$$a_{i,j} = W_{j-1} x_{i,j-1} + b_{j-1}, \qquad x_{i,j} = \sigma(a_{i,j})$$

$$\delta_{i,j} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,j}}$$

$$\delta_{i,n} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,n}} = \frac{\partial \ell(y_i, x_{i,n})}{\partial a_{i,n}} = \frac{\partial \ell_i}{\partial x_{i,n}} \frac{\partial x_{i,n}}{\partial a_{i,n}}$$

– 1986 –
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$
$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$
$$\vdots$$
$$\hat{y}_i = x_{i,n}$$
$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$

$$\frac{\partial \ell_i}{\partial W_{j-1}} \qquad \frac{\partial \ell_i}{\partial b_{j-1}}$$

$$a_{i,j} = W_{j-1} x_{i,j-1} + b_{j-1}, \qquad x_{i,j} = \sigma(a_{i,j})$$

$$\delta_{i,j} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,j}}$$

$$\delta_{i,n} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,n}} = \frac{\partial \ell(y_i, x_{i,n})}{\partial a_{i,n}} = \frac{\partial \ell_i}{\partial x_{i,n}} \frac{\partial x_{i,n}}{\partial a_{i,n}}$$
$$= \frac{\partial \ell_i}{\partial x_{i,n}} \frac{\partial \sigma(a_{i,n})}{\partial a_{i,n}}$$

– 1986 –
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$
$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$
$$\vdots$$
$$\hat{y}_i = x_{i,n}$$
$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$

$$\frac{\partial \ell_i}{\partial W_{j-1}} \qquad \frac{\partial \ell_i}{\partial b_{j-1}}$$

$$a_{i,j} = W_{j-1} x_{i,j-1} + b_{j-1}, \qquad x_{i,j} = \sigma(a_{i,j})$$

$$\delta_{i,j} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,j}}$$

$$\delta_{i,n} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,n}} = \frac{\partial \ell(y_i, x_{i,n})}{\partial a_{i,n}} = \frac{\partial \ell_i}{\partial x_{i,n}} \frac{\partial x_{i,n}}{\partial a_{i,n}}$$
$$= \frac{\partial \ell_i}{\partial x_{i,n}} \frac{\partial \sigma(a_{i,n})}{\partial a_{i,n}} = \frac{\partial \ell_i}{\partial x_{i,n}} \odot \sigma'(a_{i,b})$$

- 1986 -
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
$$x_{i,2} = \sigma\left(W_1 x_{i,1} + b_1\right)$$
$$x_{i,3} = \sigma\left(W_2 x_{i,1} + b_2\right)$$
$$\vdots$$
$$\hat{y}_i = x_{i,n}$$
$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$

$$\frac{\partial \ell_i}{\partial W_{j-1}} \qquad \frac{\partial \ell_i}{\partial b_{j-1}}$$

$$a_{i,j} = W_{j-1} x_{i,j-1} + b_{j-1}, \qquad x_{i,j} = \sigma(a_{i,j})$$

$$\delta_{i,j} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,j}} \qquad \delta_{i,n} = \frac{\partial \ell_i}{\partial x_{i,n}} \odot \sigma'(a_{i,b})$$

$$\partial a_{i,j} = W_{j-1}^{\top} \partial x_{i,j-1}$$

– 1986 –
BACK PROPAGATION

$$x_{i,1} = \texttt{entree}$$
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$$\partial a_{i,j} = W_{j-1}^{\top} \partial x_{i,j-1} \implies \frac{\partial \ell_i}{\partial x_{i,j-1}} = W_{j-1}^{\top} \frac{\partial \ell_i}{\partial a_{i,j}}$$

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$$\partial a_{i,j} = W_{j-1}^{\top} \partial x_{i,j-1} \implies \frac{\partial \ell_i}{\partial x_{i,j-1}} = W_{j-1}^{\top} \frac{\partial \ell_i}{\partial a_{i,j}}$$

$$\frac{\partial \ell_i}{\partial x_{i,j-1}} \odot \sigma'(a_{i,j}) = (W_{j-1}^{\top} \delta_{i,j}) \odot \sigma'(a_{i,j})$$

$$\delta_{i,j-1} = (W_{j-1}^{\top} \delta_{i,j}) \odot \sigma'(a_{i,j})$$

– 1986 –
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✓

$$\delta_{i,j-1} = (W_{j-1}^{\top} \delta_{i,j}) \odot \sigma'(a_{i,j})$$

✓

– 1986 –
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$$\hat{y}_i = x_{i,n}$$
$$\mathcal{L}\left(W_1, b_1, W_2, b_2, \ldots\right) = \ell\left(y, \hat{y}\right)$$

$$\boxed{\begin{gathered} \frac{\partial \ell_i}{\partial W_{j-1}} = \delta_{i,j} \ell_i x_{i,j-1}^{\top} \\ \frac{\partial \ell_i}{\partial b_{j-1}} = \delta_{i,j} \ell_i \end{gathered}}$$

$$a_{i,j} = W_{j-1} x_{i,j-1} + b_{j-1}, \qquad x_{i,j} = \sigma(a_{i,j})$$

$$\delta_{i,j} = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial a_{i,j}} \qquad \delta_{i,n} = \frac{\partial \ell_i}{\partial x_{i,n}} \odot \sigma'(a_{i,b})$$

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– 1990 –
RE:AI WINTER

– 1990 –
RE:AI WINTER

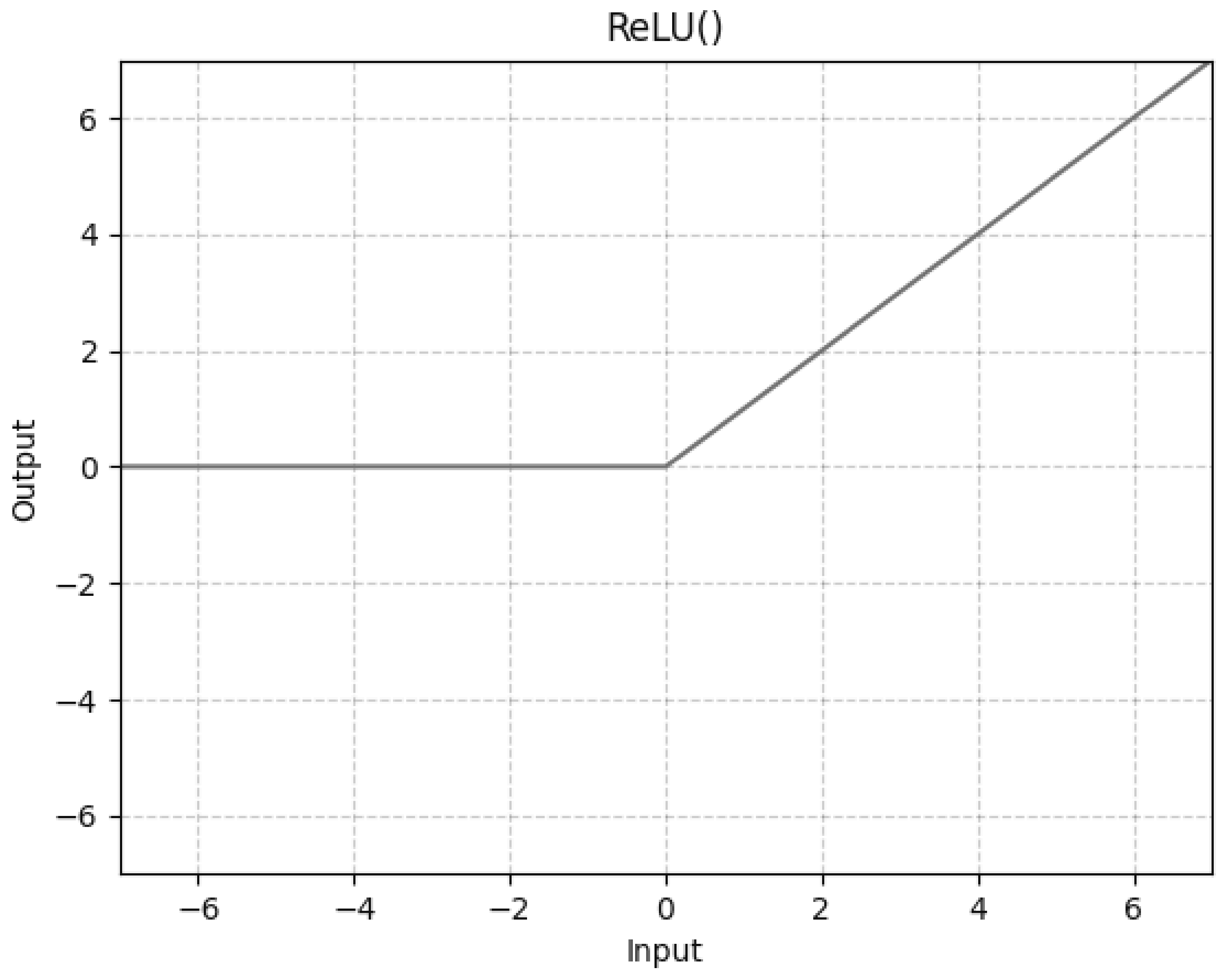
Problème du gradient évanescent

– 1990 –
RE:AI WINTER

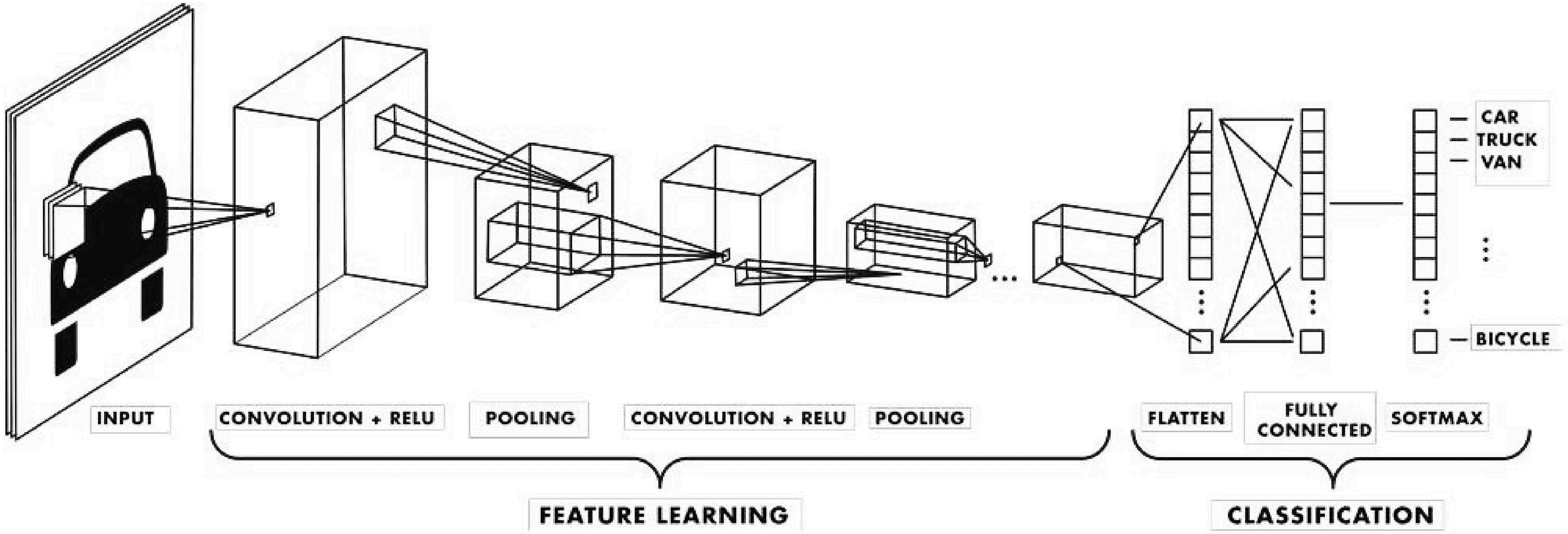
Problème du gradient évanescent

$$\sigma'(a) \leq 0,25$$

– 1990 –
RE:AI WINTER



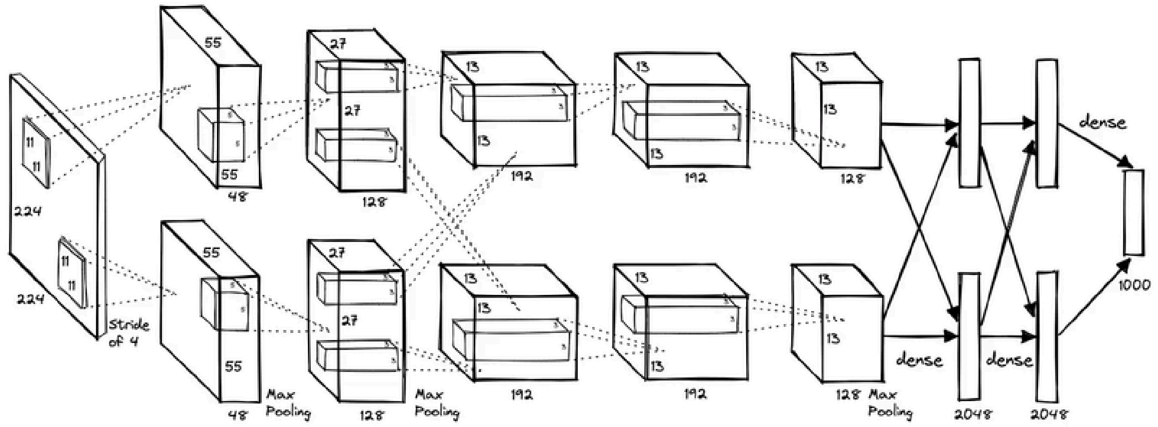
– 1990 –
CNN



Yann Le Cun

– 2006 –
RENAISSANCE

Alex Krizhevsky
Chercheur



Ilya Sutskever
Chercheur,
Confondateur d'OpenAI

[Et voilà...]

Fin de l’atelier :)

SIAHAAN--GENSOLLEN Rémy, 2026